

حكومة دبي
GOVERNMENT OF DUBAI



المجلس التنفيذي
THE EXECUTIVE COUNCIL

From Smart to Adaptive Cities:

Exploratory Benchmarking and Directional Insights for Dubai

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Prepared by
Empact Consulting



Key Definitions



Adaptive City

A city with the capacity to continuously sense change, coordinate responses across systems and adjust policies, infrastructure and services over time in response to evolving risks, constraints and opportunities.



Adaptivity

The degree and nature of a city's responsiveness over time, encompassing both real-time operational responses and longer-term structural adjustments.



Strategic Adaptivity

Longer-horizon system reconfiguration through institutional reform, spatial planning, infrastructure investment and policy redesign.



Operational Adaptivity

Real-time sensing, prediction and response within existing systems (e.g., service optimization, dashboards, AI-enabled operations).



Operating Logic

The way decisions are prioritized, coordinated and executed across government entities, markets and institutions, including the balance between central control and decentralized action.



Structural Model

The underlying configuration of governance arrangements, institutional roles, economic strategy, operating systems and incentives that shapes how a city plans, invests and responds to change.



Greenfield

The development of new physical districts, infrastructure or digital systems built from first principles, unconstrained by legacy assets, architectures or institutional arrangements.



Brownfield

The adaptation, integration, or upgrading of existing physical assets and digital systems, constrained by legacy infrastructure, data architectures, institutional processes and operational dependencies.



Transferability

The extent to which adaptive initiatives can be adapted and implemented in another city, given differences in governance structure, institutional capacity, market dynamics and development context.

Acronyms

Acronym	Full Form
AAC	Active Ageing Centres
ADIA	Abu Dhabi Investment Authority
ADQ	Abu Dhabi Developmental Holding Company
AGWA	AgriFood Growth & Water Cluster
AI	Artificial Intelligence
AIATC	Artificial Intelligence and Advanced Technology Council
ATRC	Advanced Technology Research Council
A*STAR	Agency for Science, Technology and Research
AUM	Assets Under Management
AV	Autonomous Vehicles
CCP	Career Conversion Programmes
COC	Community of Care
CNS	Community Networks for Seniors
CSF	Centre for Strategic Futures
CTP	Capability Transfer Programme
DET	Department of Economy and Tourism
DGE	Department of Government Enablement
DUCT	Digital Urban Climate Twin
EHR	Electronic Health Record
ERP	Electronic Road Pricing
EV	Electric Vehicle

Acronym	Full Form
FEC	Future Economy Council
HDB	Housing Development Board
HELM	Health, Endurance, Longevity & Medicine
IOT	Internet of Things
ITM	Industry Transformation Maps
JID	Jurong Innovation District
JTM	Job Transformation Maps
MaaS	Mobility as a Service
NTU	Nanyang Technological University
NUS	National University of Singapore
ODP	Open Digital Platform
OEM	Open Electricity Market
OTR	Operation & Technology Roadmap
PDD	Punggol Digital District
PMO	Prime Minister's Office
R&D	Research and Development
RAHS	Risk Assessment and Horizon Scanning
RIE	The Research, Innovation and Enterprise 2030 plan
SAVI	Smart & Autonomous Vehicle Industries
SGPO	Singapore Government Partnerships Office
SGTS	Singapore Government Tech Stack

Acronym	Full Form
SNDGG	Smart Nation and Digital Government Group
SNSP	Smart Nation Sensor Platform
SSG	SkillsFuture Singapore
SUTD	Singapore University of Technology and Design
SWF	Sovereign Wealth Fund
TII	Technology Innovation Institute
VCC	Virtual Commercial City
WOG	Whole of Government

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02 Part 1: Key Learnings from Structural Models – Singapore and Abu Dhabi

03 Part 2: Comparative Benchmarking of Domain-Specific Initiatives and Enablers

04 Conclusion and Directional Recommendations for Dubai



Introduction (1/2)

In recent years, disruption has become systemic, as interconnected forces, including geopolitical, economic, technological, societal and environmental, increasingly threaten and reshape stability and progress

Risk Categories



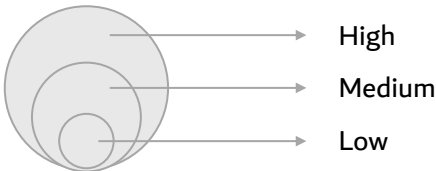
- Economic
- Environmental
- Geopolitical
- Societal
- Technological

Edges – Relative Influence

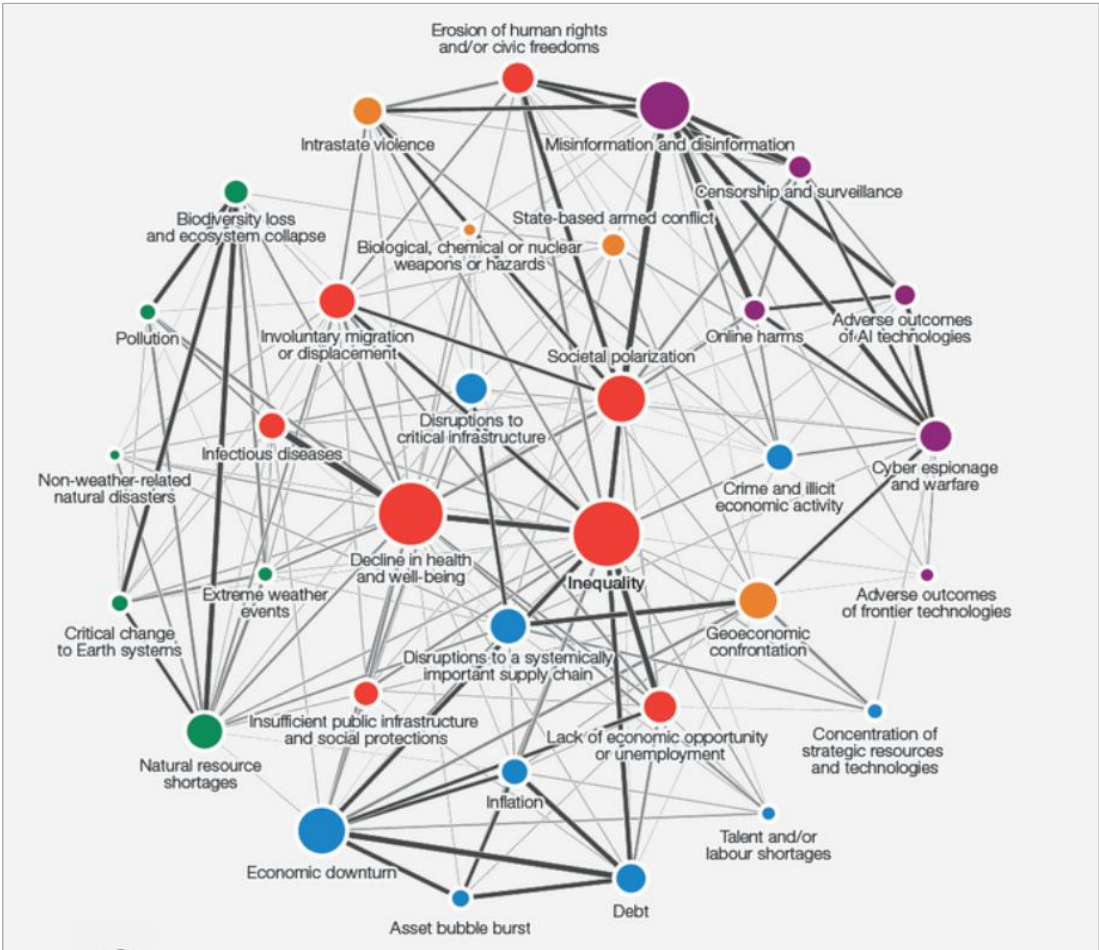


- Low
- Medium
- High

Nodes – Risk Influence



Global Risks Landscape: An Interconnections Map



- Cities worldwide face unprecedented challenges due to climate change, rapid urbanization and growing inequalities
- These issues demand innovative, collaborative solutions on a global scale
- However, most urban communities lack the integrated, forward-looking insights needed to address complex and interconnected 21st-century challenges
- Fragmented data and siloed systems hinder collaborative, proactive planning, impeding the emergence of resilient, sustainable cities capable of adapting to rapid change

*Source: WEF Global Risks Report 2025, www.adaptivecities.org

Introduction (2/2)

While the ‘smart city’ concept worked well for steady state conditions, with incremental or episodic shocks, cities must now evolve into ‘adaptive cities’ capable of responding dynamically to continuous change and disruption

An adaptive city is an urban system that can continuously sense, interpret and adapt to changing conditions, including climate change, social and economic shifts, and technological advances. It balances lives and livelihoods, resilience and sustainability and the political, economic, social, technological, legal & environmental conditions that need to be continuously analyzed and acted upon.



Smart vs Adaptive Cities*				
Smart Cities		Adaptive Cities		
What is happening now?" → Optimize current operations in real time		 Decision Logic		"What is likely to happen? How should the system respond as conditions change?" → Decisions, investments and operations across domains are continuously re-aligned in response to emerging signals, observed outcomes and evolving cross-system interactions
- Leverage sensors, connectivity and data analytics to monitor and optimize operations within sectors - Reactive and efficiency-driven, with human supervision		 What they do		- Apply predictive AI, machine learning and digital twins to forecast scenarios, to inform coordinated decisions across policy, regulation, investment and operations, enabling anticipatory responses rather than reactive optimization - Analytical and simulation tools support anticipatory responses, but adaptivity emerges from how systems change behavior under uncertainty, not from tools alone
- IoT infrastructure with sensors and connected assets - Real-time analytics and dashboards for operational visibility - Reactive optimization of traffic, utilities and public services - Rule-based automation of routine functions and alerts - Digital government portals offering transactional access - Humans retain full operational control		 Key Features		- Predictive analytics for future conditions / demand patterns and proactive optimization of resources and operations - Cross-domain data sharing and foundational interoperability - Machine learning continuously improving forecasting accuracy - Digital twin simulation for scenario planning and testing - Systems self-optimizing within human-set parameters - Human oversight for strategic and high-impact decisions

Smart and adaptive cities represent different ways urban systems respond to change. These are not sequential stages: Cities may exhibit adaptive behavior in specific domains, functions or decision contexts before such behavior becomes coordinated across the system.

*There are no universal definitions for ‘smart’, ‘adaptive’, or ‘cognitive’ cities; this classification was developed based on existing literature, to provide the foundation for assessing city maturity.

Research Objectives, Methodology and Framework (1/3)

This research report builds on a preliminary benchmarking exercise conducted by the TEC on adaptive cities...

TEC has conducted a preliminary benchmarking exercise to map relevant ‘adaptive city’ global developments in comparison to Dubai’s own initiatives and implementation models in this area. This rapid research study draws key lessons learnt from an exploratory benchmarking analysis of adaptive city approaches, enabling ecosystems, and capabilities; thus, providing a basis for further detailed research and feasibility assessment of contextually relevant initiatives for Dubai.



Research Objectives

- Conduct a benchmarking study covering adaptive-city initiatives across selected global cities
- Analyze the core enablers, governance models and cross-domain mechanisms that underpin adaptivity across urban systems
- Assess the relevance and applicability of global adaptive-city models to Dubai’s context
- Benchmark domain-specific adaptive initiatives across selected global cities and assess the key takeaways from these initiatives for Dubai
- Identify key lessons, enabling conditions and risks from global experiences to inform directional choices for Dubai across structural enablers and domain-specific adaptive initiatives



Research Methodology

- A high-level, exploratory research methodology combining research and expert consultations
 - **Desk research** focused on a review of competitiveness frameworks, city-approaches based on their historical triggers and context, strategies, governance models, implementation mechanisms, key initiatives, enablers and expansion plans
 - **Expert consultations** via Key Informant Interviews (KIIs) with domain specialists, policy experts and futurists to complement desk research and to contextualize findings, validate interpretations and identify implications most relevant to Dubai

Study Scope and Limitations

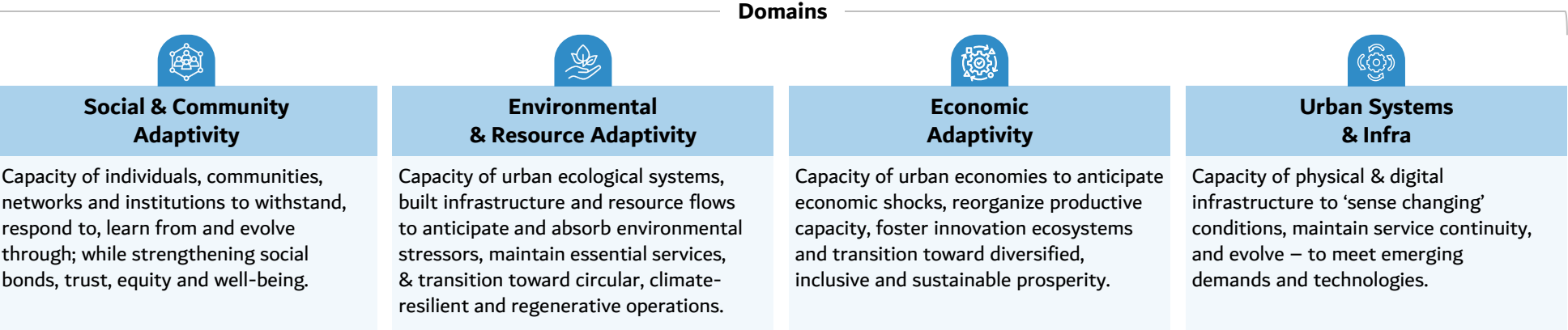
- This study is a high-level, exploratory assessment intended to surface structural patterns & directional insights, rather than provide weighed options, prioritized recommendations or implementation plans
- Findings are informed by international benchmarks and indices used as indicative signals, not as precise or fully standardized performance measures
- Data availability and comparability vary across cities and domains, with time lags and proxy indicators common in cross-city analysis
- The study does not include detailed technical design, costings or feasibility assessments for sector-specific or capital-intensive initiatives
- Recommendations focus on governance, operating-model and project interventions and are intended to guide further prioritization and analysis rather than serve as implementation directives

*Source: Dubai Municipality and SIA Partners ‘Cities’ Livability In the Age of Global Citizenship’ Report

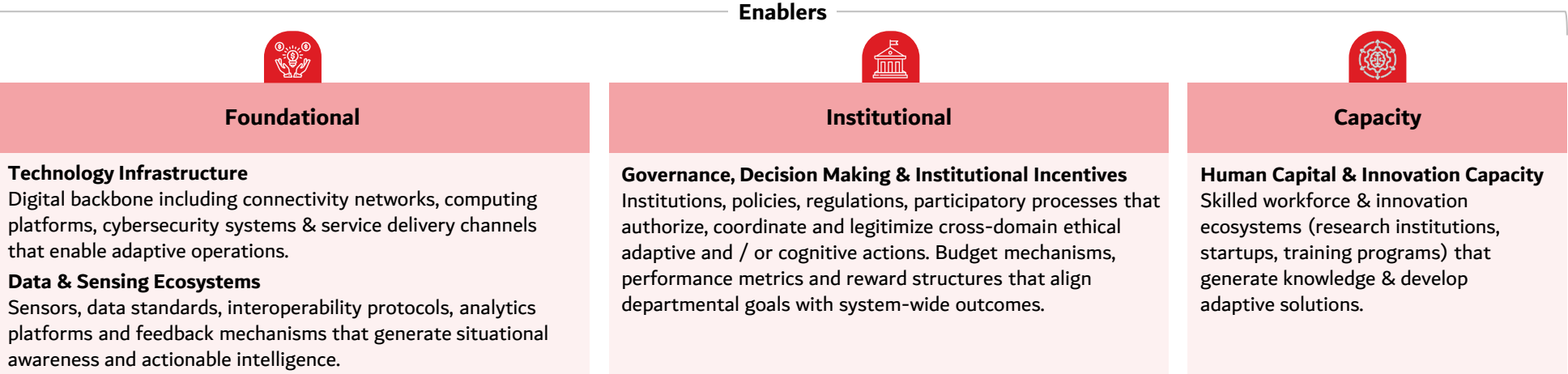
Research Objectives, Methodology and Framework (2/3)

...and undertakes a deeper, multi-city, multi-domain benchmarking exercise to assess adaptive capabilities, good practices and transferable initiatives relevant to Dubai's context

Benchmarked Domains and Enablers



Cross Domain Integration



Benchmarked Cities*

- 

Singapore
- 

Abu Dhabi
- 

Seoul
- 

Tallinn
- 

Zurich
- 

Geneva
- 

Helsinki

*All the benchmark cities were studied; however, Singapore and Abu Dhabi deeply assessed, being the most relevant comparators. The three cities (Abu Dhabi, Singapore and Dubai) share economy-led development priorities, transient populations, environmental constraints and centralized top-down governance models, alongside a strong technology-driven approach in service design.

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Research Objectives, Methodology and Framework (3/3)

The report covers (I) adaptive city models with enabling conditions, and (II) cross-country initiatives to surface practical insights for Dubai.

Part I: Structural Pathways to Adaptive City Development

Part II: Cross-domain Benchmarking



Cities



Abu Dhabi



Singapore



Abu Dhabi



Singapore



Seoul



Talinn



Zurich + Geneva



Helsinki



Rationale

Abu Dhabi and Singapore represent two distinct, high-performing models of adaptive-city development. Studying their governance architectures, digital stacks and development pathways provides structural insights that are most relevant to Dubai’s long-term trajectory.

Cities demonstrate different strengths across the four adaptive domains (economic, social, environmental, urban). Benchmarking them highlights the initiatives, enablers and system configurations underpinning their performance.



Research Questions

- How have Abu Dhabi and Singapore designed and evolved their adaptive systems and what core decisions shaped their pathways?
- How are their government stacks, governance models and data architectures organized to deliver adaptivity at scale?
- Which structural features of these models are most applicable and transferable to Dubai’s decentralized, commercially driven context?

- Which benchmark cities lead across specific adaptive domains & what differentiates their performance?
- What initiatives and enabler configurations can be attributed to their leadership?
- What domain-specific lessons and adaptive practices offer realistic and impactful opportunities for Dubai, aligned with Dubai’s D33 and the Social Agenda?

Key findings and directional recommendations

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Introduction

02

**Part 1: Key Learnings from Structural Models –
Singapore and Abu Dhabi**

03

Part 2: Comparative Benchmarking of Domain-
Specific Initiatives and Enablers

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Conclusion and Directional Recommendations
for Dubai



Part I: ‘Adaptive City’ Structural Models

This section is grounded in the premise that adaptive city outcomes are shaped by the structural models through which cities plan, govern, invest in and coordinate adaptive action

Why Structural Models Matter for Adaptive Capacity

This section examines the structural foundations that have enabled cities to develop adaptive capacity over time. A city's structural model comprises: How adaptive priorities are set, how the state intervenes, how entities coordinate, how investment is deployed, how developmental initiatives are implemented, how talent is sourced and ultimately, what ‘adaptive initiative’ responses are achievable within these parameters.

Research indicates that urban adaptiveness is fundamentally shaped by long-standing governance and institutional choices, which create both constraints and enabling conditions that persist over decades.

These structural arrangements, along with cities’ economic models and capital deployment frameworks influence:

- How quickly and effectively cities can align planning, investment and delivery across domains;
- Whether cities pursue long-horizon system change or rely on project-by-project responses and;
- Whether adaptive responses remain siloed or scale organically across the urban system

Accordingly, this section examines how the comparator cities’ structural models (the underlying configuration of governance arrangements, institutional roles, economic strategy, operating systems and incentives) shape their ability to anticipate change, adjust course and sustain adaptiveness.

It then draws out the implications for Dubai, examining how its current structural model shapes the speed, scale and coherence of adaptive responses in comparison with peer cities.

Benchmarked Cities*



Singapore



Abu Dhabi

Rationale

	
Comparability	Contrast
Economy-led growth, top-down strategic steering, transient populations, lack of urban-rural divide, reliance on FDI, focus on digitally-enabled service delivery.	Distinct drivers, state roles, governance structures and development rationale shaping adaptive capacity.

Part I: 'Adaptive City' Structural Models

Accordingly, direct comparator city models have been analyzed, focusing on the underlying structural architecture that drives their operational models and in turn, influences the 'adaptive' initiatives that these cities have adopted today



Foundational – Inherited Structural Architecture

Defines what drives adaptive priorities, how the state exercises authority, where capital originates and how entities relate to one another. These elements are deeply embedded and typically costly to change. These elements define the underlying conditions within which a city operates and adapts.

- *Primary Drivers:* Core economic, social, environmental or strategic imperatives shaping adaptive priorities
- *Role of the State:* How government steers, enables, coordinates or directly operates systems
- *Coordination Mechanisms:* How public and private entities relate to one another, including levels of integration, authority and autonomy



Operational – Delivery & Execution Model

Defines how investment is structured, how development is undertaken and how talent is sourced, deployed and retained. These elements are partially adjustable but operate within foundational constraints; they determine how structural conditions translate into action. The elements include:

- *Investment logic:* How resources are deployed, including long-horizon system building versus project-based interventions
- *Development approach:* The balance between greenfield system creation and brownfield adaptation of existing physical and digital systems
- *Digital and data enablement:* How technology, data platforms and analytics are deployed to support coordination, decision-making and service delivery
- *Talent Model:* How skills are sourced, developed and retained, given demographic and labour-market conditions



Outcomes – Achievable Adaptive Outcomes

Defines which adaptive initiatives are feasible and scalable and the level of adaptive maturity that be realistically sustained. Outcomes emerge from the interaction between foundational and operational layers. Such initiatives may:

- Have 'operational adaptivity' (capacity for real-time sensing, response and optimization within existing systems and parameters) and / or 'strategic adaptivity' (built-in flexibility for future evolution, including ability to reconfigure systems as conditions change)
- Remain sector specific or enable coordination across multiple urban systems and sectors

Part I: ‘Adaptive City’ Structural Models

The “smart city” label masks the distinct structural drivers shaping Singapore, Abu Dhabi, and Dubai. Each city’s unique constraints and endowments have produced different adaptive strengths.

	 Singapore	 Abu Dhabi	 Dubai
 Constraints / Endowments	Constraint-driven Systematization Forced independence, limited land, no natural-resource base.	Vision-driven Sovereign Investment Hydrocarbon wealth; intergenerational investment horizons.	Competition-driven Federated Agility Limited oil; dependence on external capital, labour, trade flows.
 Primary Drivers	- Sustaining long-term national resilience and competitiveness via resource maximization - Managing density, housing and economic growth within a fixed territorial footprint	- Anticipated post-hydrocarbon transition and the need to secure long-term economic resilience - Strategic ambition to position Abu Dhabi as a global leader in future sectors	- Sustaining competitiveness as a global hub for trade, capital, tourism and talent - Maintaining growth through speed, flexibility, and continuous market alignment
 Structural Response	- Centralized approaches to land use, transport, housing and utilities to manage density and efficiency - Long-term national spatial and infrastructure planning to manage competing land uses - National-scale system design to reduce duplication and fragmentation across urban domains	- Deployment of sovereign wealth funds and state-linked entities as instruments of development - Use of state capacity to manage long-horizon risks that markets alone would not absorb <i>Recent key pivots:</i> Tech and knowledge economy focus, AI native government vision, green energy, inter-government department coordination focus	- Strong emphasis on ease of doing business and investor confidence; free zones and semi-autonomous authorities accelerate sector / infra growth and service delivery - Project-led delivery and rapid deployment are prioritized “First in region” positioning carries strategic importance as it directly influences investment attraction
 Adaptivity Strengths	- High institutional coherence, with integration embedded at the planning & infrastructure layers prior to service development - Promotion of integration supports city-wide predictive platforms and autonomous anticipatory responses - Smart Nation (2014) operationalized existing integration, rather than creating it	- Capital-led adaptivity with the ability to build or reconfigure systems at scale, supported by patient funding - Demonstrated potential to strengthen resultant adaptivity through greater unification and integration, enabled by consolidated authority and a comparatively less fragmented ecosystem	- Entity autonomy delivers world class domain-specific capabilities and outcomes; differentiated service models and front-end service excellence <i>Structural trade off:</i> Federated agility enables speed but created coordination costs for cross – entity coordination

These trajectories reflect different foundational conditions and drivers (the “Why”) that drove distinct structural responses (“how the city organized itself”), which in turn determines the architecture of how each city adapts: through system-wide coordination (Singapore), capital-led system building (Abu Dhabi) or rapid market-driven experimentation (Dubai).

Part I: ‘Adaptive City’ Structural Models

These distinct structural drivers translate into different state roles and coordination mechanisms, shaping governance models and how each city is institutionally organized to adapt

	 Singapore	 Abu Dhabi	 Dubai
 Model / Key Narrative	Centralized Integration Model Trusted, high-performance global hub built on efficiency, reliability and long-term national resilience; focus on public services, productivity and global competitiveness.	Sovereign Capital Model Falcon economy’ with long-horizon capital focused on economic diversification, talent attraction, deeper government integration and frontier projects.	Federated Market Model Top global city to live, work and invest; centered on customer experience, happiness, innovation, FDI, technology and infrastructure excellence.
 State Role	<i>Coordinator + architect</i>: A highly centralized, lean government that shapes markets through regulation, long-range planning and targeted public investment	<i>Architect + investor + operator</i>: A state-capital model where Sovereign Wealth Funds (SWFs) drive sector development, R&D anchoring and long-horizon strategic bets	<i>Enabler + market-maker</i>: Government shapes business environments through regulatory design, large-scale public-private projects and strong state-linked developers that catalyze private sector growth
 Coordination Mechanisms	- Highly centralized, Whole-of-Government (WOG) architecture, enabled by governance arrangements, unified land ownership, statutory planning authority, and interoperable systems - Unified digital stack and integrated service journeys for individuals and businesses	- Movement to unified digital government with enforced standards, led by Department of Government Enablement (DGE) - Strong sovereign-backed development through Mubadala, ADQ and G42 - Consolidated service backbone portal (TAMM)	- Private-sector-driven growth, with government as regulator and enabler - Decentralized ecosystem of semi-autonomous entities and multiple free zones with high operational independence - Movement towards partnerships and ‘unification’ pilots
 Governance Model	Structural, anticipatory and system-driven	Centralized, directive and investment-led	Agile, market-responsive, experimental

These models reflect distinct structural contexts and therefore do not translate uniformly across cities. Singapore’s integrated approach is underpinned by a degree of centralized authority that is not structurally present in Dubai; Abu Dhabi’s long-horizon capital-led model reflects endowments that Dubai must generate primarily through commercial activity; and Dubai’s emphasis on speed and autonomy reflects structural conditions that support market responsiveness rather than system-wide integration.

Part I: 'Adaptive City' Structural Models

These structural models translate into distinct investment logics, delivery approaches, digital enablement strategies and talent configurations



Singapore



Investment Logic

- Long-horizon system investment via unified budgeting and statutory boards; patient capital for multi-decade infrastructure. Counter-cyclical investment capacity enables acceleration during downturns



Development Approach

- Strong emphasis on brownfield adaptation and system retrofitting, complemented by targeted greenfield development
- Extensive below-ground and integrated planning



Digital Enablement

- Access to 100+ government services with personalized recommendations and resources for users via LifeSG
- Unified national digital stack through GovTech
- CODEX framework for cross-agency data sharing
- Common platform for 40+ agencies with 200+ cloud-based systems via Singapore Government Tech Stack (SGTS)



Talent Model

- Citizen-core model with strategic foreign talent in specialist roles
- Public investment in continuous upskilling
- Tripartite cooperation manages workforce transitions
- Long-term talent planning aligned with economic strategy



Abu Dhabi

- Long-cycle SWF-directed investments aimed at economic diversification and future system creation, including de-risking priority platforms to attract private and foreign investment

- Predominantly greenfield system creation through new purpose-built districts and zones, with brownfield adaptation concentrated in digital overlays rather than physical retrofits

- Centralized digital government platform (TAMM) supports standardized service delivery and data-enabled coordination, with increasing emphasis on AI-driven optimization
- DGE enforces cross-entity standards

- Focused global talent attraction with Emiratization for strategic roles
- Deepening investment in frontier research institutions, creating talent pipelines and industrial capacities



Dubai

- Project-based and opportunity-driven deployment emphasizing speed, visibility and market responsiveness
- Selective system investments where incentives align
- Freezone model accelerates capital attraction

- Greenfield mega-project model with speed-to-market prioritized
- Entity-led development delivers strong domain performance but limits citywide coordination

- Federated digital ecosystem (DubaiNow) with strong front-end integration and entity-led data systems
- Cross-entity analytics developing incrementally through shared platforms

- Highly mobile, global workforce with shorter tenures
- Talent attraction prioritized over long-term institutional knowledge accumulation

Singapore, Abu Dhabi and Dubai often deploy comparable operational instruments, including large-scale projects, digital platforms, foreign talent and pro-investment regimes. However, these instruments are embedded within distinct structural and governance contexts, which shape these operational choices and how they translate into system-level adaptiveness.

To exemplify, Singapore's long-horizon investment, unified digital stack and national talent development initiatives translate existential constraints into system-wide responses



Source: <https://oecd-opsi.org/innovations/virtual-twin-singapore/>, <https://geospatialworld.net/prime/case-study/aec/singapore-goes-underground-with-the-digital-underground-project-2/>, <https://www.straitstimes.com/singapore/transport/erp-2-0-govt-at-very-early-stage-of-looking-at-possible-distance-charging-models-says-chee>, <https://lkyspp.nus.edu.sg/gia/article/singapore-s-traffic-success-offers-roadmap-for-region>, <https://www.pub.gov.sg/Public/WaterLoop/OurWaterStory>, <https://www.straitstimes.com/singapore/environment/to-help-indoor-vertical-farms-singapore-research-project-wants-to-optimize-crop-seeds>, <https://www.skillsfuture.gov.sg>, <https://www.wsg.gov.sg/home/about-wsg>, <https://policy.asiapacificenergy.org/sites/default/files/BCA%20Green%20Mark%20.pdf>, <https://isomer-user-content.by.gov.sg/50/7a481fd0-e545-417e-a14b-e414efee6694/rb172978-mnd-abc-water.pdf>, https://www.c40knowledgehub.org/s/article/Strategies-for-cooling-Singapore?language=en_US

City Card: Singapore

To exemplify, Singapore’s long-horizon investment, unified digital stack and national talent development initiatives translate existential constraints into system-wide responses



Structural Constraint	Adaptive Requirement	Illustrative Initiatives
Human capital as primary asset (economic competitiveness depends on workforce capability)	Continuously upgrade workforce to maintain hub economy position	SkillsFuture: A ‘Lifelong Learning’ national movement, to ensure every Singaporean (student or adult) acquires the skills required by the future economy, directly linked to future economic priorities and employment data. ‘MySkillsFuture’ is the digital one-stop portal for Singaporeans to access ‘SkillsFuture’ initiatives, find courses, use their credits to learn, explore career paths and view job opportunities. It uses AI for personalized learning pathways via platforms like CareersFinder, which uses data driven insights to provide personalized recommendations Workforce Singapore: Aims to build a competitive, resilience and inclusive workforce by supporting both individual Singaporeans and local employers in navigating economic transitions, through skilling and job redesign
Climate Risk (sea-level rise is an existential threat; Urban heat island effect leading to health risks)	Continuously strengthen and harden land, water, drainage, buildings and public infrastructure	Coastal Protection Megaprojects: Designed using Virtual Singapore scenarios – Polders (reclaimed land protected by dikes) planned for City-East area; Offshore barriers and seawalls; Integrated coastal-inland drainage systems ABC Waters Program: Transforms drains / canals into naturalized streams with real-time water level monitoring; Combines flood management with urban amenity BCA Green Mark Scheme: A green building rating system to promote sustainable construction, design and operation, especially for tropical climates: new public sector buildings must achieve “Green Mark Platinum” and 80% of new developments will be "Super Low Energy (SLE)" buildings from 2030 Cooling Singapore Programme: A catalogue of 86 heat mitigation strategies and measures available to policymakers in Singapore, grouped into seven clusters

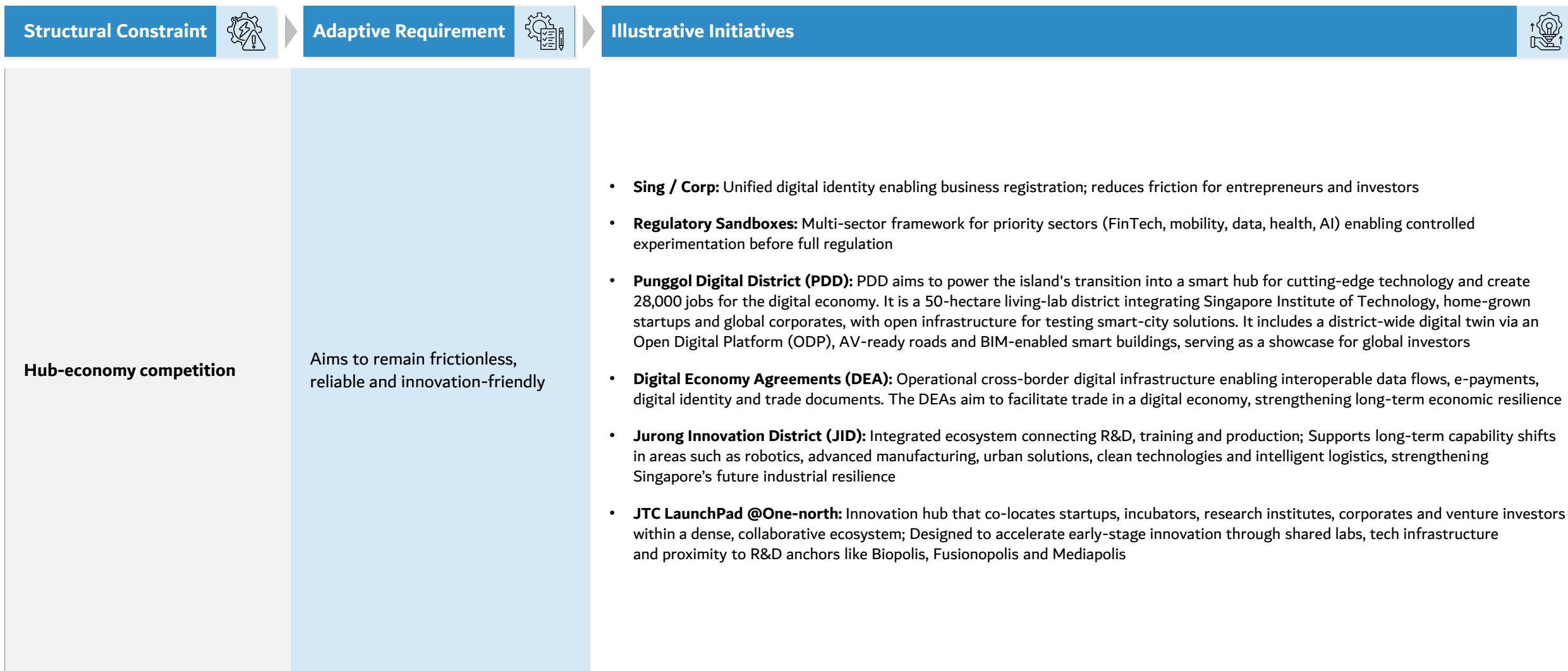
City Card: Singapore

It aims to produce integrated responses to demographic transition management and its aspiration to stay globally competitive and attract the trade, capital, talent and capabilities needed to thrive

Structural Constraint	Adaptive Requirement	Illustrative Initiatives
Ageing demographics (21% 65+ in 2024; 25% by 2030)	Mitigating rising healthcare burden; labour constraints; and providing preventive, embedded care	Smart HDB Elderly Monitoring: HDB (Housing & Development Board) operates 80% of Singapore’s public housing. It offers free Wireless Alert Alarm systems (AAS) to all seniors living in public rental blocks, provides paid commercial packages for seniors in owned HDB flats, subsidized senior-friendly fittings and Personal Alert Buttons to vulnerable individuals. At the city level, senior-centric upgrading works are in progress, with initiatives such as barrier free access, wayfinding features, fitness trails and therapeutic gardens HealthierSG: Systemic reform shifting from hospital-based treatment to GP-led preventive care; residents enroll with a family doctor who coordinates screenings, vaccinations and lifestyle interventions, reducing downstream system burden Community of Care (COC): The COC is a government-supported ecosystem focused on helping seniors and vulnerable residents age in place with dignity, bridging hospital care with home / community support through local networks, services and dedicated professionals to manage health and social needs, shifting care from hospitals to the community Health District @ Queenstown: Initiative entails neighborhood testbed redesigning estates, mobility and social ecosystems for ageing-in-place. Successful projects piloted in the health district are to be included in future rejuvenation plans and scaled to other towns and estates National Electronic Health Record: "One Patient, One Health Record" across all healthcare providers; rollout progressing in phases SG Cares + Active Aging Centres (AACs) : 24 ‘SG Cares’ Volunteer Centres’ + 223 AACs shift the state’s role from being the sole provider of care to being the enabler & platform operator. Together with Healthier SG, the AACs ensure that 8 in 10 seniors now have access to a center near their home; they offer a wide range of programmes across five key domains of social, physical health, cognitive, learning and volunteerism Community Networks for Seniors (CNS): Cross-agency data and predictive analytics identify at-risk seniors before crisis, triggering coordinated outreach across GPs, hospitals, social service agencies and community partners

City Card: Singapore

It aims to produce integrated responses to demographic transition management and its aspiration to stay globally competitive and attract the trade, capital, talent and capabilities needed to thrive



Singapore's initiatives are supported by four foundational enablers: centralized governance authority, systematic human capital development, unified technology infrastructure and integrated data ecosystems

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City Card: Singapore

Singapore's initiatives are supported by four foundational enablers: centralized governance authority, systematic human capital development, unified technology infrastructure and integrated data ecosystems



Dimension	Details	Why it Matters
Technology Infrastructure	Government Tech Stack (SGTS) & GCC 2.0: Centralized “Whole-of-Government” stack providing reusable components (NECTAR PaaS, APEX API Exchange, SHIP-HATS CI / CD); Enables agencies to build / scale apps without separate and standalone infrastructure Digital Utility Stack: Four foundational “digital utilities” for the whole economy: digital identity (Singpass and Corp), e-payments & e-invoicing (e.g., InvoiceNow), document attestation and data exchange Secure Network & Infrastructure Standards: Unified cybersecurity controls and guardrails (e.g., SHIP-HATS, MAESTRO) ensure consistent security, observability and compliance across agencies	- Agencies build on shared infrastructure rather than standalone systems - Common rails reduce duplication and enable interoperability across government and economy
Data & Sensing Ecosystems	Smart Nation Sensor Platform (SNSP): Smart Nation Sensor Platform (SNSP) with nationwide sensor network (including plan to convert 110,000 lampposts into interconnected wireless sensors) Virtual Singapore: High-resolution 3D national digital twin combining real-time and legacy data from multiple agencies to simulate urban scenarios (e.g., infrastructure, environment) for planning and resilience Open & Operational Data Platforms: data.gov.sg and SG’s Digital Utility Stack (e.g., APEX, MyInfo, SGFinDex) act as shared data utilities for government and financial sectors	- Real-time situational awareness supports anticipatory governance - Agencies can access cross-domain data without relying on manual requests, allowing planning decisions to be informed by integrated, current information rather than siloed snapshots

City Card: Singapore

These responses are enabled by structural conditions specific to Singapore's context, that may vary from Dubai's federated governance model



Structural Differences

Conditions specific to Singapore's city-state model

	Singapore	Dubai
Jurisdiction	Unitary government controls all policy levers and domains, with no federal coordination required.	Operates within UAE federal structure; cross-emirate coordination may add further complexity.
Population stability and continuity	Stable citizen base (~69% citizens and Permanent Residents) with multi generational stake in social integration and community outcomes.	~92% expatriates, with historically shorter tenure (6.7 years), now averaging 10+ years; limits longitudinal investment in population.
State land ownership / Public housing platform	The 99-year lease is the fundamental model for land tenure; HDB provides direct channel to 80% of population.	Extensive freehold property rights with expatriates in private / rental housing; no equivalent platform for population-scale intervention.

Transferable with Adaptation

Policy choices that could be selectively adopted

	Singapore	Dubai Relevance
Existential resource framing and fiscal insulation for long-horizon investments	Water, energy and food framed as existential threats, justifying long-term planning horizons and financial commitments that standard cost-benefit analysis would not support.	Singapore demonstrates specific mechanisms that existential framing unlocks: ring-fenced funds protected from annual budget cycles, access to national reserves under "life-threatening" criteria, inter-generational borrowing instruments and legislative codification ensuring continuity These mechanisms address long-horizon, cross-system risks that typically fall outside the mandate, incentives or time horizons of individual entities..
Sectoral Strategic Planning linked with human capital attraction / retention	JTMs explicitly link industry transformation to workforce skills, training pathways and career progressions, creating end-to-end coordination from economic strategy to employment outcomes.	A model for connecting sector strategies to talent pipelines within a single planning instrument.
Enforcement Authority	Digital planning and execution support consolidated under the highest office, enabling cross domain coordination.	Selective placement and consolidation of authority at the highest levels of government to strengthen enforcement in priority domains within Dubai's jurisdiction.
Unified Service Integration Layer	NRIC provides universal identifier; Singpass enables authentication across 2,700+ services; LifeSG consolidates 100+ citizen services into single interface with 1.5M+ users.	A model for emirate-level service consolidation built on existing federal identity infrastructure, supporting proactive anticipatory governance.
Research-to-Deployment Integration	R&D outcomes are deliberately pulled into deployment through coordinated testbeds and scale-up pathways.	Adoption of mechanisms for integrating experimental outcomes into operational systems.

City Card: Abu Dhabi

Abu Dhabi's state-led transformation targets the post-oil transition through sovereign capability-building, AI-driven initiatives, preventive healthcare, and investment in climate-resilient, knowledge-intensive sectors



Structural Constraint	Adaptive Requirement	Illustrative Initiatives
Oil-dependent economy (target 64% non-oil by 2030, currently contributes 56.8% to non-oil GDP)	Diversify into knowledge-intensive sectors; build industrial base beyond extraction.	KEZAD (Khalifa Economic Zone): Adaptive industrial ecosystem with "build-to-suit" facilities and flexible zoning for shifting demand ADNOC Downstream (TA'ZIZ & Fertiglobe): Builds capacity for blue / low-carbon ammonia production and hydrogen value chains; Demonstrated early exports, support long-term energy and climate strategy via clean fuels and industrial transition ADQ Industrial Clusters: State-led consolidation building sovereign capability in pharmaceuticals, advanced materials, EV manufacturing Smart Manufacturing Program: Industry 4.0 adoption via grants, advisory and Smart Manufacturing Index
Small national population	Build sovereign technology capability that does not depend on imported expertise.	Technology Innovation Institute (TII): Advanced R&D in AI, quantum computing and advanced materials; developed Falcon 180B (sovereign LLM) Stargate UAE: One of the largest sovereign AI compute deployments globally, enabling AI training, simulation and cross-sector deployment Quantinuum–TII Collaboration: Early-stage quantum computing for energy, logistics, finance SAVI (Smart & Autonomous Vehicle Industries): Multi-modal AV cluster (air, land, sea) for R&D, testing, certification
Healthcare system pressures (lifestyle diseases, rising expectations, national population health)	Shift from treatment to prevention; integrate health data for proactive intervention.; build sovereign clinical AI.	Malaffi: Unified health information exchange connecting all public and private providers; Foundation for AI-supported decision-making Emirate Genome Programme: 850,000+ Emirati genomes sequenced, with AI-powered analysis for early intervention Universal Newborn Genetic Screening: 20+ metabolic / genetic conditions screened at birth Smart Healthcare Platform + Sahatna Portal: Unified appointments, diagnostics, AI triage, telehealth, personalised alerts Med42 (Clinical LLM): Abu Dhabi-developed medical AI for triage, diagnostics, clinical documentation

City Card: Abu Dhabi

Abu Dhabi's state-led transformation targets the post-oil transition through sovereign capability-building, AI-driven initiatives, preventive healthcare, and investment in climate-resilient, knowledge-intensive sectors



Structural Constraint	Adaptive Requirement	Illustrative Initiatives
Extreme climate and resource scarcity (50°C+ temperatures; minimal rainfall; coastal exposure)	Secure water and food supply; adapt operations to heat; protect coastline; monitor environmental change.	- AGWA (AgriFood Growth & Water Cluster): Alternative proteins, algae-based feed, reverse osmosis, smart logistics - Controlled-Environment Agriculture: High-yield vertical farms optimising water, nutrients, yield under extreme conditions - Mangrove Restoration Programme (100M by 2030): Coastal protection, carbon sequestration, biodiversity - EAD Environmental Data Platform: 30 years of data; AI / ML predictive modelling and early warning - Heat-Risk Early Warning & Worker Protection: Microclimate sensors; Automatic work-rest triggers integrated with ADNOC HSE
Hub-economy competition (vs Dubai, Singapore, Doha for talent, capital, innovation)	Remain attractive to global innovators while building local ecosystem.	- Hub71: Founder-centric ecosystem with flexible capital, incentives, ADGM access - HELM (Health, Endurance, Longevity & Medicine) at Masdar City: Life sciences cluster with 28-day clinical trial approval - Ma'an – Social Innovation: Incubates mental health, disability inclusion, youth wellbeing solutions via private / non-profit partnerships - DCT Smart Tourism: AI-enhanced visitor system for real-time guidance and cultural mediation
Establishing frontier-state differentiation	Position as global testbed for cognitive urbanism and AI-integrated government.	- Abu Dhabi Digital Strategy 2025-2027: AED 13B investment to become world's first fully AI-native government by 2027; 100% sovereign cloud adoption; 200+ AI solutions across services - TAMM 4.0: World's first "AutoGov" function; 1,100+ services from 90+ partners; Predictive delivery, AI Vision, multilingual assistance. - Aion Sentia (planned 2027): World's first cognitive city; Learns and adapts to residents via MAIA AI platform integrating mobility, energy, healthcare, housing "AI for All" Program: 95% of 30,000+ government employees completed AI training; citizen upskilling for digital participation

City Card: Abu Dhabi

Abu Dhabi's initiatives are supported by four foundational enablers: patient sovereign capital, recent efforts to vertically integrate governance which links leadership to execution....



Dimension	Details	Why it Matters
Governance, Decision-Making & Institutional Incentives	Unification of Government Services and Capacities: Department of Government Enablement (DGE) is the central entity responsible for enabling Abu Dhabi Government to operate as “One Government”. This includes building shared digital infrastructure, talent management programmes, governance and policy standards. All government agencies are subject to the oversight and supervision of the DGE. The DGE Chairman sits on Executive Council, ensuring a direct line from strategic direction to implementation Dedicated AI / Technology Institutional Architecture Advanced Technology Research Council (ATRC): Established in 2020 to define R&D priorities across academia and industry; Houses the Technology Innovation Institute (TII) for applied research, ASPIRE (technology programmes) and VentureOne (commercialization) Artificial Intelligence and Advanced Technology Council (AIATC): Established in 2024 to develop AI policy, strategy and investment coordination MGX: Established in 2024 as a dedicated vehicle for AI infrastructure, semiconductors and core technologies Vertically Integrated R&D-to-Deployment Pipeline: TII (research) → VentureOne (commercialisation) → G42 / Core42 (deployment infrastructure) → MGX (scale investment) Sovereign Investment Architecture (~\$1.7T+ assets under management across Abu Dhabi) Abu Dhabi Investment Authority (ADIA): Long-term global portfolio investor; Diversified returns for intergenerational savings; Active in AI / data center co-investments Mubadala Investment Company: Strategic investor in technology, aerospace, healthcare, semiconductors; key shareholder in G42, GlobalFoundries; Highly active dealmaker (\$29B across 52 deals in 2024) Abu Dhabi Developmental Holding Company (ADQ): Domestic critical infrastructure and strategic assets (Etihad Airways, AD Ports Group, Pure Health, TAQA) MGX (launched March 2024 by AIATC): Dedicated vehicle for AI infrastructure, semiconductors and core technologies with Mubadala and G42 as founding partners; Led \$40B “Aligned Data Centres” deal with BlackRock / Global Infrastructure Partners G42: Technology holding company (Mubadala-backed, with \$1.5B infused investment from Microsoft in 2024); Houses Core42 (sovereign cloud and AI infrastructure), M42 (healthcare technology), Khazna Data Centres, Presight (analytics) International Holding Company (IHC) and related entities under Royal Group: Additional technology and strategic investments, including stakes in G42	- Leadership overlap eliminates translation gaps between vision and implementation - ATRC consolidates R&D strategy, investment and commercialization pathways under single authority - AIATC ensures AI treated as strategic national priority with dedicated policy coordination - Ecosystem ensures the benefits of integrated “R&D to implementation” cycle - Differentiated capital pools for differentiated purposes enable multi-decade investment horizons that commercial capital cannot sustain

City Card: Abu Dhabi

...strategic technology partnerships and purpose-built data infrastructure



Dimension	Details	Why it Matters
Human Capital & Innovation Capacity	Public sector capacity development - DGE oversees capability development and leadership training for Government workforce (previously undertaken by Abu Dhabi School of Government); “AI for All” programme trained 95% of 30,000+ government employees in AI fundamentals Emirati Talent Pipeline for Priority Sectors - Specialized universities (MBZUAI, Khalifa University, UAEU) focus on research fields aligned with the emirate’s strategic clusters - AI and autonomous systems (SAVI), agri-food and water security (AGWA), energy, aerospace and advanced manufacturing; Supports local talent and capability-building in priority sectors - ATRC’s NexTech Program places exceptional STEM students in leading global universities - DGE’s AI Majalis builds public AI literacy through traditional community forums; Brings together leaders, innovators and experts to foster collaboration, accelerate AI adoption and align with Abu Dhabi’s vision to become the world’s first AI-native government by 2027; Features discussions, tech showcases and programs driving the UAE’s digital future Sovereign-led and cluster-based innovation pipelines: State entities (Mubadala, ADQ) and specialised zones (SAVI, AGWA, Masdar, Hub71) drive coordinated R&D, investment, testbeds and talent development around priority sectors Specialized Visa & Talent Attraction: E.g., Golden Visa, Create Visas, “Thrive in Abu Dhabi” program attract specialised international expertise	- DGE ensures government workforce can specify, deploy and oversee advanced systems - University-cluster alignment creates deliberate talent pipelines into priority sectors - Sovereign-led pipelines integrate R&D, commercialization and talent development under coordinated strategy - Public literacy (AI Majalis) builds social license and reduces adoption friction
Technology Infrastructure	Sovereign Government Cloud: Migration of Abu Dhabi government systems to a sovereign cloud platform (with Microsoft, G42); Hosts TAMM and government digital services securely in-country AI Compute Infrastructure: Stargate UAE 1-GW cluster (largest outside US) – G42 / Khazna-built, OpenAI / Oracle-operated and featuring NVIDIA Grace Blackwell systems Unified TAMM Platform: TAMM 4.0 provides an AI-enabled front end and shared orchestration layer for 1,100+ services across 30+ entities, consolidating service delivery onto a single platform Strategic Partnerships: Microsoft (\$1.5B in G42); NVIDIA; OpenAI; Oracle; Cerebras provides access to frontier models and hardware Shared ERP / finance & HR platforms: DGE is rolling out unified finance, HR and procurement systems and a government financial capability programme to standardize back-office processes across entities	- Government data remains under Abu Dhabi jurisdiction while accessing global-standard capabilities - Compute capacity at scale required for frontier AI training and inference, without dependence on overseas data centers - Eliminates routine administrative burden, freeing capacity for higher-value functions
Data & Sensing Ecosystems	Sectoral Data Platforms: Includes Malaffi (health), Environmental Data Platform and other platforms, providing unified datasets for supervision and system-wide analytics Abu Dhabi Open Data Platform: Aggregates datasets from 20+ entities (data.abudhabi) to support innovation, transparency and economic development LivAI Digital Twin: DMT platform providing real-time livability scores by district and 3D predictive modelling for urban planning	- Domain-specific platforms convert raw data into decision-ready intelligence across health, environment, urban systems - Integrated platforms provide cross-domain visibility for coordinated urban management

City Card: Abu Dhabi

These initiatives are enabled by structural conditions specific to Abu Dhabi's context, that may vary from Dubai's market-driven model

Structural Differences

Conditions specific to Abu Dhabi (Abu Dhabi)

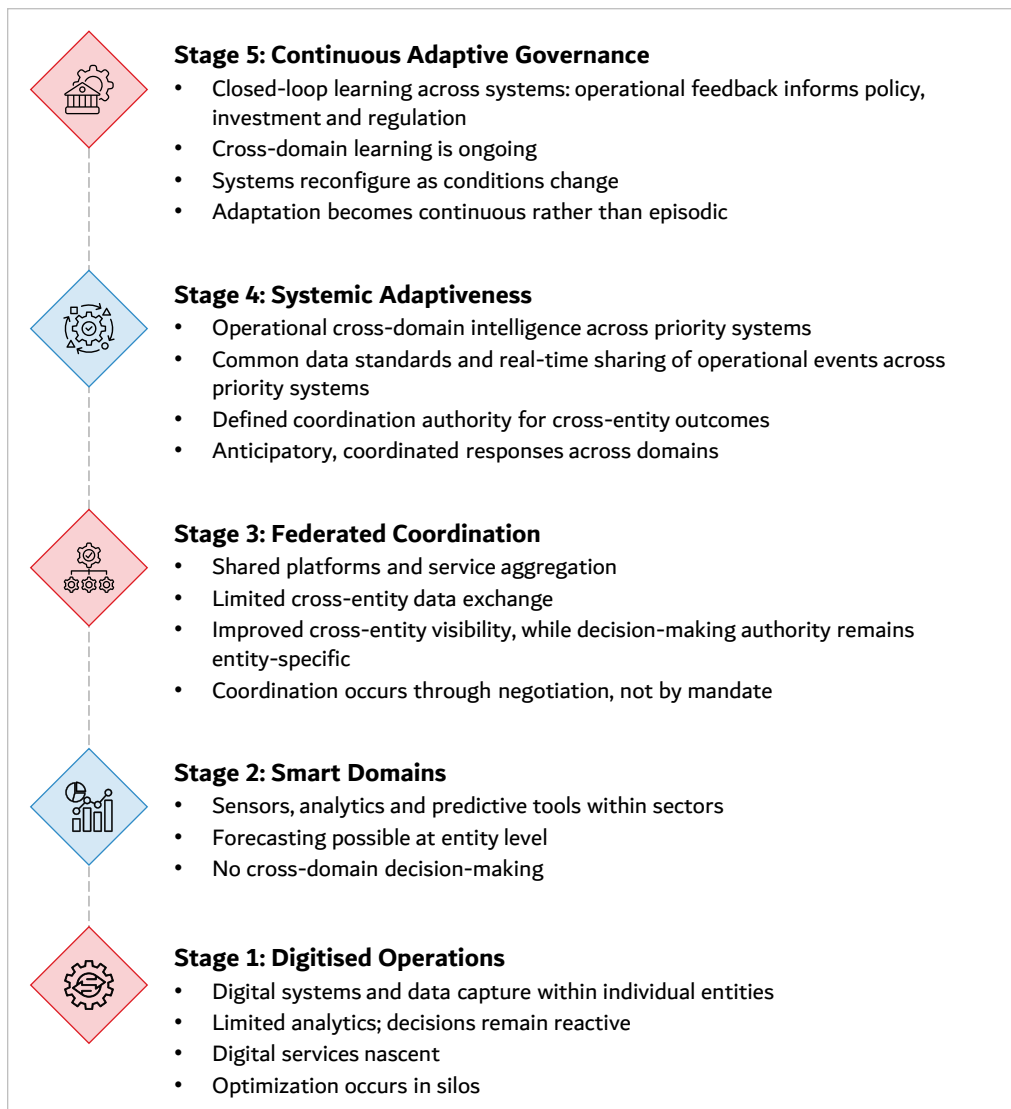
	 Abu Dhabi	 Dubai
Resource endowment	Oil reserves generate continuous surplus and patient capital for strategic investment.	Minimal oil reserves; Growth driven primarily by commercial economy.
Investment structure	~\$1.7T AUM across differentiated SWFs targeting global diversification, strategic sectors and frontier technology.	Smaller investment base with investment vehicles primarily focused on commercial returns.
Economic diversification stage and strategy	State-led diversification; Sovereign entities drive priority sectoral investments.	Market-driven growth through commercial attractiveness, connectivity and free zone competition.
Coordination Architecture	Historically centralized structure with fewer autonomous entities and free zones; Conducive to unified policy implementation.	Multiple powerful semi-autonomous entities and competing free zones; Coordination across distributed authority.

Key Implications for Dubai

- The UAE pursues a coordinated national agenda on FDI, talent and priority sectors. Within this framework, various emirates have developed distinct operating models shaped by different structural economic contexts
- Abu Dhabi has advanced large-scale initiatives enabled by patient sovereign capital and governance arrangements that can absorb long development cycles and course corrections (e.g., Masdar City launched in 2006, significantly scaled back from its original zero-carbon ambitions and is now repositioned as a “clean tech hub”)
- Dubai, by contrast, operates within a commercially driven, expat-led hub economy, where competitiveness depends on speed, regulatory responsiveness, investor confidence and visible outcomes
 - This has favored project-led innovation, rapid approvals, strong FDI attraction and high-visibility development
 - Dubai’s operating model emphasizes a “citizen-as-customer” approach: adaptive capacity is expressed through seamless front-end experiences and ease of doing business, rather than comprehensive backend integration across government
 - Applied innovation is primarily driven by sector champions and specialized free zones, supported by enabling institutions, flexible regulation and sandbox mechanisms. This structure supports experimentation and market responsiveness, but limits “whole-of-city” coordination
- However, Dubai realizes the value of the importance of integration for data and anticipatory governance.
 - Dubai has already been pursuing strategic unification where the commercial case is clear (Unified License, NABIDH in health, digital twin partnerships between Dubai Municipality and Dubai Land)
- These differences reflect structural trade-offs, not ambition gaps: Abu Dhabi’s strengths lie in long-horizon capital and directed coordination; Dubai’s in speed, commercial discipline, private-sector leverage and global connectivity

Adaptive Operational Maturity Stages

The structural models examined shape how the three cities' (Singapore, Abu Dhabi and Dubai) urban systems sense conditions, share information and coordinate responses – placing them at different stages of adaptive maturity



City	Stage	Key Characteristics
 Singapore	Stage 4 with elements of Stage 5	• Cross-domain intelligence, coordination authority and execution institutionally embedded • Operational feedback informs standards, land use and long-term investment • Real time sensors; predictive AI across 3+ domains; strong cross-domain data integration; digital twin used to simulate solutions before implementation • <i>Trajectory.</i> Core urban systems already integrated; further gains depend on managing trade-offs across on resolving cross-domain trade-offs rather than adding new coordination mechanisms
 Abu Dhabi	Stage 3 with elements of Stage 4	• Deep operational adaptiveness in sectors, wherever intentionally designed • Unified digital government, robust governance authority, early predictive AI • Uneven adaptiveness across the full city system • <i>Trajectory.</i> Pursuing AI-native government transformation and the development of a cognitive city 'Aion Sentia'
 Dubai	Stage 3	• High operational intelligence and optimization at entity and sector level • Strong front-end integration, regulatory agility and market responsiveness • Cross-domain insights exist but are not routinely converted into coordinated action • Decision authority remains distributed across semi-autonomous entities • <i>Strength.</i> Strategic adaptiveness (rapid economic and policy pivots) • <i>Constraint.</i> Limited system-wide operational coordination

Structural Choice for Dubai's Adaptive Path

Dubai's adaptive capacity is shaped by the interaction between its structural enablers and structural constraints. The trajectory of its future adaptiveness depends on how deliberately these trade-offs are recognized and managed

Dubai's Structure Enables	Dubai's Structure Constrains
 Rapid execution and visible delivery of initiatives Customer-centric service innovation Strong private-sector leverage and sustained FDI attraction Competitive dynamics between entities driving improvement Experimentation and piloting through sector champions and free zones Commercial discipline, with projects required to demonstrate viability Flexibility to pivot policy and investment direction as conditions evolve	 Long-horizon investments lacking a clear commercial or market rationale Deep backend integration across all government entities and free zone authorities State-directed technology development and frontier R&D at scale "Whole-of-Government" coordination Anticipatory governance requiring unified data models and decision rights Patient capital for non-commercial system formation Sustained population-level interventions within a largely transient demographic base

Adaptive Trajectory Options			
Option	What it Means	Pros	Cons / Trade-offs
A. Optimize Within the Current Structure (Near Term)	Continue prioritizing speed, experimentation & sector-led delivery within existing governance arrangements	- Preserves execution speed, flexibility and investor confidence - Enables rapid piloting and strong domain-level performance - Minimal governance disruption - builds on proven strengths	- City-wide, anticipatory adaptiveness remains limited - Cross-domain trade-offs are managed informally or ex post - System learning remains fragmented - Coordination depends on alignment and negotiation, rather than mandate
B. Selective Coordination Build-out (Accelerated Current Trajectory)	Formalize and scale what is already emerging: targeted coordination where system risk is highest, without "Whole-of-Government" centralization	- Improves cross-domain learning and trade-off management in priority domains - Retains sector agility while raising adaptive capacity where risks are highest - Institutionally feasible within current structures	- Uneven adaptiveness across the city - Requires disciplined scoping to avoid coordination sprawl - Benefits accrue incrementally rather than uniformly
C. Extend Coordination Capacity (Long Term)	Introduce clear mandates and decision rights for cross-domain outcomes across major urban systems	- Enables anticipatory governance, system-level learning and coordinated responses to compound shocks - Raises ceiling of adaptive capacity beyond sectoral optimization	- Requires governance reform and clarity on authority - Some loss of speed in discretionary decisions - Risk of perceived centralization if mandates not well defined

Dubai has repeatedly demonstrated the capacity to reorient its growth drivers, regulatory frameworks and sectoral priorities in response to shifting economic conditions. This strategic agility remains a defining strength. As urban challenges become increasingly interconnected and governance shifts globally toward more anticipatory approaches, the key policy question is – “whether operational adaptiveness should deepen to match this strategic capability, or whether domain-level excellence within a decentralized structure will suffice”

01	Introduction
02	Part 1: Key Learnings from Structural Models – Singapore and Abu Dhabi
03	Part 2: Comparative Benchmarking of Domain-Specific Initiatives and Enablers
04	Conclusion and Directional Recommendations for Dubai

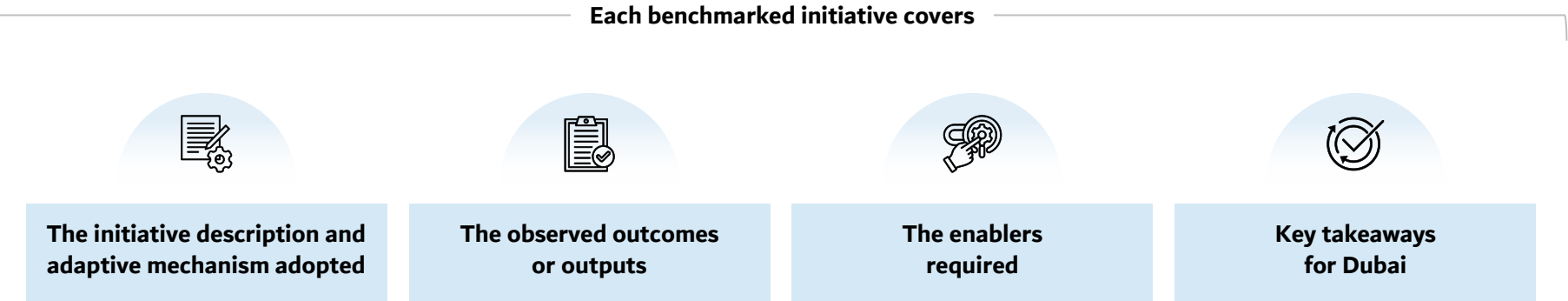
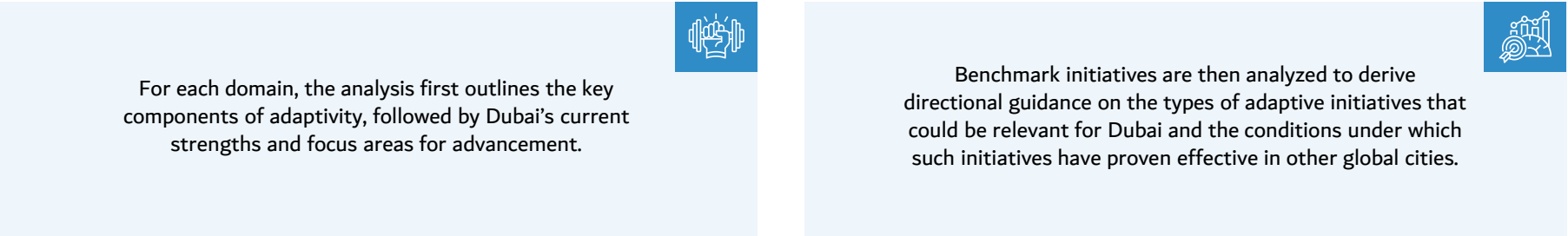


Part II: Comparative Benchmarking of Domain-Specific Initiatives

This section examines specific adaptive initiatives within key urban domains, drawing on global benchmarks to understand how adaptivity is operationalized in practice at the initiative level

Domain Specific Initiatives

The focus of this section is on concrete interventions that demonstrate anticipatory response, learning and adjustment within defined domains.



Benchmarks assessed are not intended for direct replication. Instead, they are used to derive Dubai-specific takeaways that indicate what learnings may be relevant; the enabling conditions they require; and what further assessment would be needed to test their feasibility and impact in relation to Dubai’s context.

Benchmarked Cities*

- 

Singapore
- 

Abu Dhabi
- 

Seoul
- 

Tallinn
- 

Zurich
- 

Geneva
- 

Helsinki

*Not all cities are referenced in every domain; benchmarks are applied selectively based on relevance.

City Performance Across Indices – Domain Excellence

The performance on global indices establishes a baseline, revealing specific areas where global cities / peers derive their specific strengths towards adaptivity in specific domains

						
City Ranks	IMD Smart City Index 2025	IESE Cities in Motion 2024	Kearney Global Cities Resilience Index 2025	Arcadis Sustainable Cities Index 2024	Mercer Quality of Living 2024	Global Start Up Ecosystem (GSER) '25
 Singapore	9	6 (#2 Asia)	8	18	28 (1 st in Asia)	9
 Abu Dhabi	5	116	13	Not Ranked	85	Not in top 40
 Helsinki	11	26	Not Assessed	Not Ranked	26	Not in top 40
 Zurich	1	12	Not Assessed	21	1	33
 Seoul	13	13	Not Assessed	11	81	8
 Geneva	3	40	Not Assessed	37	3	Not in top 40
 Tallinn	22	72	Not Assessed	Not Ranked	86	Not in top 40
 Dubai	4	81 (#2 regional)	4	Not ranked in top 100	83	Not in top 40 (2nd in MENA)

- **Singapore is the most relevant benchmark for state-led, integrated adaptive development.** Its consistent top performance across the IMD Smart City Index, IESE Cities in Motion Index and the Kearney Global Cities Index reflects a highly coordinated model built on national R&D capability, unified digital infrastructure and long-horizon planning
- **Seoul provides a high-value reference for economic adaptivity and operational intelligence.** Its top-ten global startup ecosystem ranking, strength in knowledge and funding dimensions and progress in sustainability indices, make it a leading model for innovation-driven growth
- **Zurich represents the global archetype for balanced, high-trust adaptive performance.** Its consistent top rankings in IMD and Mercer, combined with strong IESE sustainability scores, reflect a mature model where technology, environmental management and social cohesion are integrated through long-term, consensus-driven governance
- **Helsinki is a leading benchmark for human-centric and environmental adaptivity.** It scores strongly on IESE environment and social cohesion indicators, driven by participatory governance and a structural commitment to circular-economy innovation

- **Abu Dhabi and Dubai show similar index performance patterns, with:**
 - **Strong performance on technology readiness, digital government and institutional capacity.** Notably, Dubai’s social cohesion and community participation policy was explicitly recognized in global resilience assessments
 - Both cities score relatively lower than their peers on indicators related to **environmental sustainability, human capital depth, cost of living and infrastructure intensity**

*The selected indices are city specific and together provide a balanced view across economy, technology and innovation, urban infrastructure and mobility, governance, environment, human capital, social cohesion and lived experience.

Overview of Dubai's Key Strategic Goals

Dubai's Strategic Objectives serve as contextual reference points for discussing the implications of global benchmark results

D33 Agenda



ECONOMIC AGENDA

Goals: Double economy size, rank among world's top 3 global cities.

Focus Areas: FDI, foreign trade, domestic demand, private sector investment.

100 TRANSFORMATIONAL PROJECTS

(1) Scale 400 start-ups and build 30 global unicorns, (2) Attract the world's best universities, (3) Raise the value added of the industrial sector and promote exports (4) Integrate Emiratis into the private sector (5) Make Dubai a hub for skilled workers (5) Focus on logistics, finance, manufacturing and tourism (5) Hub for incubating innovations through Sandbox Dubai.

Social Agenda



(1) Happy, cohesive, tolerant families connected to local values and national identity, (2) Efficient healthcare promoting healthy lifestyles (3) Future-ready education system enhancing human capital (4) Proactive social protection and empowerment (5) Best living experience, residential services, cultural and sporting activities.

D33 Economic Agenda is highly project-driven with tangible outcomes, a commercial orientation and an FDI attraction focus.

D33 Social Agenda primarily articulated through service reform objectives and policy frameworks rather than through discrete flagship projects.

Supporting Strategies & Initiatives



Industrial Strategy

- Increase manufacturing output and value-addition
- Enhance knowledge and innovation
- Dubai as the preferred manufacturing platform for global businesses
- 75 projects identified

Autonomous Transport

- Transform 25% of total transportation in Dubai to autonomous mode by 2030

Blockchain Strategy

- First city fully powered by blockchain technology

Metaverse Strategy

- Attract metaverse companies
- Create 40,000 virtual jobs by 2030
- Contribute approximately \$4 billion to economy

Urban Master Plan 2040

- '20-minute city' flagship project
- 5 urban centers
- Expand green spaces and beaches
- Affordable housing for Emiratis and skilled expats

Community Participation Policy

- Aims to increase community participation in shaping policy, legislation and government services
- Aims to improve the quality of services by aligning them through community needs and expectations, to be identified via public engagement mechanisms

Services 360 Policy

- Mandates a move from reactive to proactive government services delivery

Quality of Life Strategy

- Under development

Clean Energy 2050

- 75% clean energy sources by 2050

Demand Side Management Strategy

- 75% Demand-side management: 30% savings by 2030, 50% by 2050 in electricity, water, fuel consumption, compared to business as usual

Digital Strategy

- World-class infrastructure
- Unified digital platform and common enablers
- Customer-focused, integrated, easy and fast digital services
- Enhanced digital capabilities & skills

Food Security Strategy

- Diversify sources and enhance local production

Integrated Waste Management Strategy

- Innovation in waste management, recycling and energy conversion

Economic Adaptivity

Dubai's Economic Agenda is driven by its D33 Plan and sector specific strategies

Drawing on a review of leading global indices, this sub-section defines the core components of “economic adaptivity”, assesses Dubai’s current strengths and highlights the strategic direction embedded in the city’s existing policies. Together, these components form the analytical foundation for selecting relevant benchmark initiatives and international practices that can inform Dubai’s future adaptive pathways.



Component	 Productive Capacity Flexibility	 Innovation Pipeline & Renewal	 Opportunity Mobility	 Global Positioning	 Shock Detection & Response
	Ability to reconfigure production factors in response to technology or market shifts.	Capacity to generate knowledge, technologies and business models that drive economic regeneration.	Ability of individuals and firms to transition between and access growth pathways.	Ability to attract resources and talent, project influence; and integrate into global value chains	Capability to anticipate economic disruptions, coordinate responses, and institutionalize learning.
Strengths	- Diversified, growing economy driven by trade, logistics, finance; strong focus on tech solutions and digital economy - Industrial Strategy prioritizes advanced manufacturing - Flexible labour market with global talent access and streamlined visas - Agile regulatory environment with sandboxes, fast licensing and full foreign ownership; 'Unified License' launched (25)	- Strong regional innovation leadership (GSER '25 #2 in MENA) \$2.3 billion in venture capital in '25, marking a 35% increase from '24 - Robust innovation hubs, sandboxes and digital economy initiatives - Clear D33 focus on R&D, creative economy and future technologies - Strength in Fintech and Blockchain	- Strong SME base; significant entrepreneurship activity - Low unemployment; major youth employment and Emiratization initiatives - Established pathways for global talent attraction, with Dubai ranking first globally in ease of entry for talent (Kearney Global City Index)	- Top-tier global competitiveness and a world leader in attracting greenfield FDI - Powerful global brand in tourism, aviation, infrastructure and safety - Emerging tech hub positioning, with high AI vibrancy and a stated goal to be a Top 3 global city	- Proven rapid crisis response capability (COVID rebound) - Advanced digital government and smart city data foundation - Diversification strategy strengthens macro-resilience
Focus Areas For Advancement	- Limited depth in complex, high-value manufacturing and local supply chains - Early-stage adoption of Industry 4.0 - SME adaptation pressures from uneven digital maturity, legacy systems and cultural resistance - Limited evidence of integration between SME and large manufacturing enterprise integration, despite government procurement support	- R&D intensity and deep-tech commercialization remain below global front-runners - Underdeveloped university-industry research clusters and a maturing "lab-to-market" commercialization pipeline - Weaker in human capital (talent) and outputs - SME scaling to mid-cap / global champions remains limited (Dubai Founders HQ has recently been launched)	- Limited upward mobility for low-skilled expatriate workforce - Labor mobility into higher-productivity sectors, remains uneven for both nationals and long-term residents - Fragmented lifelong learning infrastructure	- Intense competition for talent retention, against other global and Gulf hubs - Limited high-value export sophistication; reliance on tech deployment versus creation activities - Regulatory coordination requirements due to the multiplicity of free zone and mainland systems. The D33 "Unified License" is a key initiative designed to address this	- Limited integrated early-warning mechanisms for economic shocks across sectors - Scenario planning not deeply integrated with economic policy cycles

Economic Adaptivity Benchmarking – Initiative Project Cards (1/6)

Singapore demonstrates how long-horizon sector transformation enables sustained productive capacity flexibility through its ITMs and OTRS...




Component	Productive Capacity Flexibility 	Focus Area	Transformation planning blueprint and toolkit 
Country	 Singapore		
Initiative	Industry Transformation Maps (ITM)		Operation & Technology Roadmap (OTR)
Description	- A national framework comprising 23 sector-specific Industry Transformation Maps, ranging from manufacturing, trade & connectivity and modern services to built environment & lifestyle. The ITMs: - Set 3 to 5-year strategies for each sector, around the pillars of productivity, skills development, innovation and internationalization - Provide guidance on how industries / market players can remain relevant, resilient and competitive in a fast-changing global landscape		- OTRs help businesses identify specific technologies and operational changes required to remain competitive within the context of the relevant industry transformation roadmap. This also leads to creation of bespoke innovation plans for select companies. The OTRs: - Enable SMEs to develop long-term strategies, plans for tech and manpower needs and build industry-relevant future capabilities - Incorporate processes for updating and reviewing roadmaps to account for business and technology changes
Impact	- Labour productivity reported to have grown 2.4% annually since ITIM roll-out - Annual reported growth of 1.9% in real median gross monthly income of full-time employed residents in the 23 ITM sectors		- Provide SMEs with a customized, expert-guided roadmap for technology adoption and innovation - Supported KoolLogix's goal of 80x growth and valuation of \$1B by 2026 - Skyfy Tech grew revenue by >175 times in 2 years and achieved a net worth of S\$3.5M
Enablers	Governance: Collaborative model led by the Future Economy Council (FEC), involving deep partnerships between government agencies, industry players, unions and education institutions Data & Foresight Capability: Ministry of Trade and Industry analyses global shifts, sector competitiveness, productivity data and labour-market needs to inform ITM updates; Skills Future & WSG (Workforce Singapore) monitor emerging skills demand Incentives & Delivery Instruments: Targeted grants, regulatory refinements and skills frameworks		Governance: Run by national R&D agency (A*STAR), lending it significant technical credibility Human Capital: Relies on expert facilitators who guide SME leadership teams through the structured, multi-workshop sensing and planning process Institutional Incentives: Part of the "GET-Up" (Growing Enterprises through Technology Upgrade) program, acting as the strategic entry point for SMEs to access A*STAR's other resources & talent

- 
 - Adaptive Mechanism:** Singapore’s ITM-OTR framework embeds anticipatory economic adaptivity by translating long-term national priorities into sector-specific capability roadmaps, workforce needs and firm-level technology pathways. This reduces the lag between emerging economic shifts and coordinated industry response.
 - Dubai Relevance:** Dubai’s D33 targets depend not only on attracting new investment, but also on raising productivity and capability depth within existing sectors. The benchmark illustrates how long-horizon sector roadmaps can complement growth ambitions by systematically strengthening firm-level productivity, workforce readiness and technology uptake over time.
 - Structural implications for Dubai:** Fully replicating Singapore’s integrated transformation architecture would imply a higher degree of centralized coordination than applicable within Dubai’s current operating model. However, Dubai could pursue similar outcomes in priority industries, through an industry-led model, with key government agencies (e.g., Department of Economy and Tourism) convening integrated roadmaps that link skills, technology adoption and SME support.

Component	Productive Capacity Flexibility 		Focus Area	Creating and placing a future ready workforce 
Country	 Singapore			
Initiative	Job Transformation Maps (JTM)	SkillsFuture Singapore (SSG) – Creating Talent	Workforce Singapore (WSG) – Placing Talent	
Description	- JTMs provide deep-dive analysis into how the changes outlined in the ITMs (such as digitalization and automation) will impact specific job roles over a 3-to-5-year horizon - This linkage ensures that economic growth (ITM) is supported by firm-level capability (OTR) and a future-ready workforce (JTM)	- A "Lifelong Learning" National Movement, to ensure every Singaporean (student or adult) acquires the skills required by the future economy, directly linked to the JTMs - SSG manages the "Skills Frameworks" (what to learn) and issues "SkillsFuture Credits" (money to learn it) directly to citizens' accounts to pay for approved courses	Career Conversion Programmes (CCP) – Helps businesses reskill mid-career talent into growth roles while offering salary support and training subsidies through the CCP grant Capability Transfer Programme (CTP) – Facilitates transfer of capabilities from foreign specialists to locals for all 23 ITM sectors	
Impact	- Provides a common reference for government agencies, employers, training providers and workers on future role demand and skills requirements - Serves as the analytical foundation for downstream programmes, including SSG and Career Conversion Programmes (WSG) * <u>Note:</u> JTMs themselves do not deliver placements; they enable coordinated action.	1.05 million Singaporeans (approx. 37% of the workforce) have utilized their credits since inception - Over 24,000 enterprises sent employees for training in 2024, with 95% being SMEs - The SkillsFuture Career Transition Programme achieved a 54% placement rate for mid-career switchers within 6 months of graduation	CCP: Between 2018 and 2023, ~37,000 workers were successfully reskilled and placed into new jobs via CCPs; Retention rates for CCP hires are 3.9 percentage points higher than standard hires CTP: Benefited >140 companies and facilitated knowledge transfer to > 1,000 local professionals across 20+ sectors.; Top 3 sectors in terms of local trainees are precision engineering, air transport and logistics	
Enablers	Governance and Institutional Integration: Led by the WSG, in collaboration with government agencies and industry; Sets the basis for a formal linkage of JTMs to SSG training frameworks and WSG programmes; ensuring insights get operationalized Data & Foresight Capabilities: Ongoing analysis of sector trends, technology adoption and global labour shifts	Incentives: A direct-to-citizen subsidy wallet Partnerships: Leveraging industry leaders to curate training	Financial Support: Subsidized wages during the training period removes the financial risk for employers hiring inexperienced mid-career staff Human Capital: Enabling pathways for transferring foreign expertise (CTP) to local talent	



- Adaptive Mechanism:** Singapore’s JTMs function as an anticipatory labour-market intelligence layer, translating sector transformation priorities into role-level job impact. By linking JTMs to Skills Future (skills creation) and Workforce Singapore (talent placement), the system aligns workforce development and redeployment in anticipation of structural labour shifts.
- Dubai Relevance:** With the average tenure of Dubai’s residents increasing over time, clearer signals on future job roles and skills demand can support both Emirati workforce development and the attraction and retention of global talent aligned with Dubai’s evolving economic priorities.
- Structural implications for Dubai:** The predominance of expatriate labour, high employer autonomy and decentralized hiring may limit the state’s ability to steer job transformation in a coordinated way.

Economic Adaptivity Benchmarking – Initiative Project Cards (3/6)

Singapore and Seoul demonstrate how SME capability upgrading can be embedded within enterprise ecosystems

Component	Opportunity Mobility 	Focus Area	SME – enterprise ecosystem integration 
Country	 Singapore	 Seoul	
Initiative	SkillsFuture Queen Bee (QB) Networks		Win-Win Smart Factory Support Program
Description	- The government (SkillsFuture SG) appoints and incentivizes large, industry-leading companies (e.g., Siemens, Microsoft, ST Logistics, PwC) to act as "Queen Bees" - These QBs then take responsibility for mentoring, providing skills advisory and co-developing training for the SMEs in their own supply chain and ecosystem		- A unique "Matching Grant" partnership between the Government (Ministry of SMEs), Large Conglomerates (Samsung, Hyundai, LG, POSCO) and SMEs. - The Large Corporation contributes 30% of the funding and deploys its own retired master engineers to the SME's factory floor for around 6 to 8 weeks to redesign processes
Impact	- Solves the "last mile" problem of SME training and scales skills development by embedding it within commercial relationships - As of 2025, 37 QBs had supported over 5,770 enterprises (80% SMEs) - E.g., PWC Queen Bee program involves a three-month "mentorship project" to tackle a top sustainability concern - E.g., QB Bosch Rexroth helped 14 SMEs customize 19 "proof-of-concepts"		- Participating SMEs recorded an average, 30% increase in productivity and a 43% reduction in defect rates - Over 2,620 SMEs were upgraded in the first 3 years alone
Enablers	Governance: Each SME that joins a QB network is given access to a "Skills Manager" who acts as the dedicated liaison. QBs often structure their support around specific, measurable projects. The QB / network tracks the program's success through both quantitative and qualitative metrics. Financing: A mix of subsidies, including Enhanced Training Support for SMEs (ETSS) (up to 90% subsidy) and Absentee Payroll (paying the SME for the time their staff is away) Institutional Incentive: Leveraging the buyer-supplier relationship		Institutional Incentive: The government publicly ranks conglomerates on their level of support to SMEs. High scorers get tax incentives, exemption from antitrust audits and public procurement preference Human Capital: Leverages a massive, underused asset, hundreds of "retired" engineers (aged 60+) from Samsung / Hyundai who have world-class expertise, but no active role Financing: Funding is gated by "Levels" (1–5). An SME cannot apply for AI funding (Level 5) until a mentor certifies they have mastered basic digital data collection (Level 1)



- Adaptive Mechanism:** The Singapore and Seoul models embed SME capability upgrading directly into buyer–supplier and enterprise relationships, enabling skills transfer, technology adoption and productivity improvement as part of regular commercial activity rather than through stand-alone training or grant programmes.
- Dubai Relevance:** SMEs account for a large share of Dubai’s economic activity, but capability upgrading is uneven and often incremental. The benchmark highlights how anchoring SME development within enterprise ecosystems can accelerate opportunity mobility by linking SMEs to real demand, technical expertise and higher-value production pathways.
- Structural implications for Dubai:** While Dubai has demand-side tools such as ICV (In country Value programme) and Tasharok that incentivize local participation; these mechanisms are primarily procurement and compliance-driven. Dubai’s SME landscape is characterized by high churn, a predominantly expatriate workforce and limited long-term buyer-supplier commitments, which constrain the systematic transfer of capabilities. This limits the effectiveness of ecosystem-based models unless commercial incentives and risk-sharing mechanisms are explicitly designed to align large firms, SMEs and government objectives (e.g., linking Emiratization outcomes or mandated capability transfer requirements to priority sectors).

Component	Opportunity Mobility 	Focus Area	Mobility based on tech and market shifts 
Country	 Tallinn		 Singapore
Initiative	e-Residency		Macro-Talent Portfolio Framework
Description	- A government-issued transnational digital identity that allows anyone, anywhere, to access Estonia's public e-services and establish and manage a location-independent, EU-based company 100% online - It decouples the nation's economy from its physical geography		- A portfolio approach that allows Singapore to be highly resilient and adaptive to global talent shocks. If one "asset class" (e.g., foreign talent) becomes too expensive or politically sensitive, the government can "rebalance" by increasing investment in another (e.g., local upskilling) - Approach leverages flexible pathways for attracting global talent e.g., Overseas Networks & Expertise (ONE) (holder is not tied to one company); Tech. (issued to attract established / mature global tech talent); and Tech@SG (flexible, fast-track employment passes for fast-growing tech companies) - Portfolio model engages diaspora for effective integration. E.g., The Singapore Global Network runs events and manages networks that produce tangible economic value, such as helping Singaporeans abroad find jobs, connect with peers, or get "practical advice on how to set up an office" in a new market - Reprioritization and rebalancing are typically achieved through upskilling the local workforce via SkillsFuture initiatives
Impact	- The program generated €68 million in direct economic impact in the first half of 2025 alone, primarily from tax revenue - The total direct impact since inception is over €342 million. It has attracted over 128,000 e-residents who have founded over 36,000 new Estonian companies		
Enablers	Technology Infrastructure (non-negotiable): The program runs on Estonia's 20+ year-old mature digital backbone: a universal e-ID system and the X-Road secure data exchange layer Governance & Decision Making: A "willingness to change," a legal framework that makes digital signatures legally equivalent to handwritten ones and a "once-only" principle for data capture Institutional Incentives: A highly competitive tax system (0% corporate tax on reinvested profits) and minimal bureaucracy		Governance & Decision Making: A highly sophisticated, data-driven civil service that views talent as a unified macro-economic portfolio, not as a siloed development pillar Data & Sensing Ecosystems: Relies on a deep, real-time understanding of skills gaps and workforce needs, supported by a national skills taxonomy



- Adaptive Mechanism:** Tallinn decouples business formation from physical geography through a mature digital backbone (X-Road, universal e-ID, legal equivalence of digital signatures). Singapore manages talent as a macroeconomic portfolio, flexibly rebalancing among foreign attraction, diaspora engagement and local upskilling, as conditions shift.
- Dubai Relevance:** Dubai already offers comparable front-end products, e.g., Virtual Commercial City (VCC) – although VCC remains constrained by fragmented backend systems (non-guaranteed banking access or digital identity). For talent, attraction and retention levers (visa policy, Emiratization, employer incentives) operate through separate channels, without a unified view of demand prioritization. Best-practice insights underscore the importance of treating mobility as a system-level enabler; linking enterprise formation, talent attraction and workforce development in a cohesive manner to effectively support the D33 priority sectors.
- Structural implications for Dubai:** Achieving this requires deeper system interoperability and policy linkage rather than new visa or licensing products. Specifically, a unified data-exchange backbone for business mobility and clearer institutional linkages between global talent inflows and Emirati capability development could be considered.

Economic Adaptivity Benchmarking – Initiative Project Cards (5/6)

Innovation pipelines in each city reflect their specific economic needs, risk profiles and strengths, rather than a convergent “best practice” model

Component	Innovation Pipeline 	Focus Area	Innovation Prioritization 
Structural Condition	What it enables	Illustrative Examples	
Existential constraint driving long-horizon state planning	Multi-decade R&D programmes tied to national survival priorities; counter-cyclical investment discipline.	Singapore: Eight successive 5-year national R&D plans since 1991, with government allocation increasing from S\$2 billion (1991-1995) to S\$37 billion (2026-2030). Public R&D investment maintained at ~1% of GDP for over 20 years, sustained across economic cycles, further attracting private investment. E.g., S\$18 billion in semiconductor industry investment secured 2023-2024.	
Sovereign capital with risk-absorption capacity	Tolerance for longer payback periods; opportunistic deployment in frontier technologies and priority sectors.	Abu Dhabi: SWFs provide patient capital for strategic sector bets and frontier technologies. E.g., (1) Semiconductors: Mubadala created GlobalFoundries (2009), now one of world's top foundries. (2) AI: MGX vehicle (launched 2024) co-led \$30 billion AI infrastructure partnership with Microsoft, BlackRock, NVIDIA – took positions in OpenAI, Anthropic, Databricks.	
Corporate-industrial base + state coordination	Open innovation platforms; university-industry clusters; public-private matching investment.	Seoul: Large conglomerates operate open innovation platforms that invest in, partner with and / or acquire startups. For example, Hyundai invested ~\$1 billion in 200+ startups (2017-2023). Government amplifies efforts by matching with public R&D grants (catalyzing \$10.7 billion in follow-on funding over 10 years).The combination creates absorption pathways (acquisitions, joint ventures, procurement contracts) unavailable in ecosystems without dominant industrial anchors.	
World-class research institutions + strong IP regime	University-anchored commercialization; structured spinoff pathways; high survival rates.	Zurich: ETH Zurich consistently ranks among the world's top 10 universities for science and engineering, producing research with commercial application. Combined with Switzerland's strong IP protection regime, this creates a pipeline for university spinoffs. For example, ETH's Transfer Office has supported 615 spinoffs since 1973, with 93% surviving beyond 5 years and CHF 425 million in private investment in 2024 alone.	
Small, high-trust society enabling regulatory experimentation	Real-world piloting environments; first customer/reference for startups; validation before international scaling.	Helsinki: Finland adopted "experimentation and deregulation" as an explicit national strategy (2015), resulting, amongst other outcomes, in the world's most permissive AV and drone testing laws. Testbed Helsinki offers city infrastructure for mobility, EdTech and climate pilots. Companies gain real-world validation, then scale internationally.	
Digital-native public infrastructure	Frictionless business formation attracting entrepreneurs; government as platform.	Tallinn: e-Residency programme has issued digital IDs to 117,000+ people from 185 countries, enabling formation of 33,800 Estonian companies (1 in 5 of all new companies in 24) The programme resulted in a €274 million cumulative direct economic impact with 10:1 government ROI.	



- Adaptive Mechanism:** Across benchmark cities, innovation pipelines are adaptive not because of individual programmes, but because they leverage their structural conditions to continuously redirect investment, experimentation and talent in response to the city’s evolving economic performance.
- Dubai Relevance:** Dubai's innovation model is based on its structural strength and is commercially-driven: regulatory agility, sandbox positioning, global brand and FDI attraction. This configuration is aligned with applied innovation and scaling, vis-à-vis frontier R&D. As regional competition intensifies, greater clarity on where Dubai seeks to lead, versus where it partners, becomes increasingly important.
- Structural implications for Dubai:** Dubai already deploys sovereign investment, regulatory sandboxes and talent programmes across priority sectors. The key insight from best practice mapping is the value of integrating these levers around sector-specific outcomes rather than operating them independently. This also means maintaining regulatory speed as competitors close the gap and strengthening retention mechanisms to capture long-term value.

Economic Adaptivity Benchmarking – Initiative Project Cards (6/6)

Singapore links short, medium and long-term foresight into a unified decision loop that anticipates and prepared for disruptions

Component	Shock Detection & Response 		Focus Area	Anticipation and Preparation 
Country	 Singapore			
Initiative	Pulse of the Economy (POE)		Risk Assessment and Horizon Scanning (RAHS)	Centre for Strategic Futures (CSF)
Description	- A dashboard used by Ministers to track economic health in real-time, bypassing the 3-month lag of GDP reports - It uses high-frequency, non-traditional data (e.g., electricity consumption, public transport data, online job listings) to get real-time insights into the economy, moving beyond quarterly GDP reports		- A “Whole-of-Government” operating system that uses AI and cognitive modeling to scan for “weak signals” of future disruptions (economic, social, security) before they become mainstream trends - Unlike standard risk management, RAHS focuses on low-probability, high-impact “Black Swan” events, allowing agencies to “pre-experience” shocks through simulation	- A dedicated “internal contrarian” unit within the Prime Minister’s Office responsible for challenging current policy assumptions - Runs “National Scenarios” exercises that force every ministry to stress-test their 10-year strategies against wild-card economic and geopolitical shocks
Impact	- During COVID-19, POE tracking allowed the government to calibrate support packages surgically (e.g., identifying which sectors were recovering vs. stalling week-by-week) - Lends the effect on shifting economic management from “Post-Mortem” (fixing what happened last quarter) to “Preventative” (fixing what is happening today)		Successfully identified shifting global trade patterns prior to major crises, allowing Singapore’s Sovereign Wealth Funds (GIC / Temasek) to adjust asset allocations early	Directly informed the “Industry Transformation Maps” (ITMs) by identifying long-term automation risks to the labor market years in advance
Enablers	Governance: The Public Sector Governance Act (PSGA) that legally enables data sharing between government agencies Human Capital: A dedicated GovTech team that builds pipelines to ingest raw data from utility companies and transport operators Infrastructure: APEX (API Exchange); Trusted Centres (Govt.-Sanctioned “Safe Rooms” for data)		Tech Infrastructure: The RAHS software engine, which aggregates global open-source intelligence to map threat clusters Network: A global network of “Future Fellows” and private sector nodes that feed diverse data into the government	Governance: CSF sits within the Strategy Group in the Prime Minister’s Office, which drives “whole-of-Government” strategic planning and influences strategic resource allocation. This institutional positioning connects foresight outputs directly to budget and policy decisions



- Adaptive Mechanism:** Singapore integrates three foresight layers: real-time economic sensing (Pulse), horizon scanning for emerging risks (RAHS) and long-term scenario planning (CSF). These operate as complementary capabilities within a “Whole-of-Government” foresight ecosystem, with CSF’s institutional position in the Prime Minister’s Office connecting foresight outputs to strategic resource allocation.
- Dubai Relevance:** Dubai’s fast-moving, externally exposed economy would benefit from earlier detection of sector-specific stress signals and structured anticipation of future disruptions, particularly as global capital flows, trade patterns and technologies shift rapidly. While Dubai Pulse is a key initiative in this direction, the platform primarily serves as a data aggregation layer rather than an analytical engine for weak-signal detection or scenario-driven planning.
- Structural implications for Dubai:** Approximating Singapore’s model would require three elements: (1) an AI-enabled analytical layer within Dubai Pulse for early risk and weak-signal detection, (2) clearer institutional ownership of horizon scanning and scenario planning functions and (3) formal linkage between foresight outputs and budget or policy decisions.

Social Adaptivity

Dubai's Social Agenda is driven by Dubai Social Agenda 2033 and the UAE Wellbeing Strategy

Drawing on a review of leading global indices, this sub-section defines the core components of “social adaptivity”, assesses Dubai’s current strengths and highlights the strategic direction embedded in the city’s existing policies. Together, these components form the analytical foundation for selecting relevant benchmark initiatives and international practices that can inform Dubai’s future adaptive pathways.

	 Life-Journey Guidance	 Social Cohesion, Belonging & Participation	 Anticipatory Care & Vulnerability Support	 Adaptive Social Infrastructure & Community Ecosystems
Component	Adaptive, personalized support across major life transitions (birth, schooling, skills, work, family formation, aging)	Ensuring residents feel anchored, connected and able to participate fully in community and civic life	Early identification and proactive support for elderly, disabled, low-income, at-risk families, and people in crisis	Evolving physical and digital community spaces that strengthen wellbeing, connection, and collective action
Strengths	- Aggregated government services and simplified access across touch-points, provided via Dubai Now platform - The Services 360' policy aims to achieve 100% proactive and automated services	- Extremely diverse, safe and globally attractive city with strong community stability indicators - Social Agenda emphasizes family resilience, cultural identity and cohesive neighborhoods - Community Participation Policy (launched '25) - Strong public trust, safety and digital engagement - support belonging - CDA's volunteering platform has ~59k volunteers	- Robust support systems for people of determination, child protection and elder care via CDA; additionally, the 'Thukkher' Card, can be used by the elderly to avail various support services - Health infrastructure is mature (100% hospital connectivity via NABIDH); prevention logic is active (EJADAH Value-based model covers 30 chronic diseases) - Social Agenda commitments for family stability and proactive protection	- High-quality urban amenities, parks, waterfronts, community centers, libraries and cultural assets - Smart city infrastructure enables real-time service improvement and citizen feedback - Legal basis for engagement, created via the Community Participation Policy
Focus Areas For Advancement	- Personalized, anticipatory guidance around transitions (early childhood → jobs → aging) - Integration of health, education, labour and social data for proactive interventions - Adult re-skilling and active aging pathways	- Inclusion of low-income, migrant and long-term expatriate communities in cultural and civic life - Structured neighborhood-level programmes that build local identity and everyday cohesion - Multilingual, culturally-adaptive service delivery	- Real-time home/utility data integration to predict social isolation or risks - Integration between health, social care, policing and community networks	- More distributed, walkable, mixed-use community hubs aligned with rising population densities - Greater use of digital platforms for hyper-local engagement, volunteering and community activation - Flexible spaces that adapt for youth, seniors, creative use and emergency response

Social Adaptivity Benchmarking – Initiative Project Cards (1/5)

Singapore demonstrates the most mature model for citizen-life journey adaptivity

Component	Life-Journey Adaptivity 		Focus Areas	Proactive Guidance 
Country	 Singapore	 Abu Dhabi		 Helsinki
Initiative	LifeSG (Operational)	TAMM (Version 4.0 announced Oct '25)		Aurora AI (Discontinued)
Description	- An application that consolidates 100+ government services and information, organized around a citizen's life's major events rather than by government agency - Includes a unified dashboard displaying eligible government benefits, offers personalized recommendations, a personal profile that pulls data from multiple agencies and an inbox for application status updates and appointment reminders	- AI-powered unified digital ecosystem, consolidating >1,100 services from 90+ government and private sector entities - Automatically manages recurring services, with users having the ability to set preferences and customize automation - New features include a conversational TAMM AI Assistant, AI Vision and Smart Guide		- Ambitious national AI program (2020-2022) designed to reorient Finland's public and private services around a "human-centric" and "ethically sustainable" model - Vision extended beyond a single app, to a decentralized open network where AI would proactively suggest services to citizens from different providers based on their specific "life events" (e.g., moving to a new city, unemployment, etc.)
Impact	Shifted government service delivery from an agency-centric, transactional model to a citizen-centric, holistic one	Supports the aspiration to create the world's first AI-native government by 2027		The program was discontinued – it was highly experimental, and its main legacy was on “key lessons learned” from its failure
Enablers	Tech Infrastructure: Enabled by Singpass (the National Digital Identity) for secure biometric authentication and APEX (API Exchange), a government-wide API gateway that facilitates secure data sharing between agencies Data Integration & Interoperability: Enabled by the Myinfo data platform and APEX (government API exchange) enabling a "Tell us once" policy Governance: Governed by SNDGO and implemented by GovTech; Operates on a strict, explicit user consent model	Tech Infrastructure: Powered by Microsoft Azure OpenAI Service and G42's Compass 2.0 enterprise AI platform; Utilizes JAIS (Arabic LLM), built using agentic frameworks like LangChain and LangGraph Data Integration & Interoperability: Enabled through a central platform that brokers service requests across government entities Governance: Launched and managed by the Department of Government Enablement		Governance: Driven by the Ministry of Finance and supported by an Ethics Board. The core concept was described by researchers as "difficult to comprehend" Data & Sensing Ecosystem: The vision was to use a "MyData" approach integrated with Finland's digital ID, giving users full control over their own data. However, the project struggled to move from this abstract ethical principle to a practical, trusted data-sharing model



- Adaptive Mechanism:** Organizing government services around citizen life events rather than agency structures, enabling proactive delivery based on where a person is in their life journey; rather than requiring them to navigate to the right agency.
- Dubai Relevance:** Dubai has made significant progress in digitizing individual services and simplifying access, but most services still operate as discrete transactions rather than connected journeys. As population complexity increases and expectations rise, the ability to coordinate services across entities around key life events becomes critical to service quality, trust and administrative efficiency. This is aligned with the ‘Services 360’ framework, whose foundation is based on a “Whole-of-Government” service policy aimed at delivering seamless, end-to-end experiences across entities.
- Structural implications for Dubai:** Life-journey adaptivity requires shared identity, reusable personal data and service orchestration that allows actions taken in one part of government to automatically trigger related services elsewhere. This depends far more on backend interoperability, consent-based data reuse and clear authority for cross-entity orchestration than on front-end applications alone. Without these structural elements, citizen services remain reactive and fragmented regardless of digital maturity.

Component	Social Cohesion, Belonging & Participation 	Focus Areas	Community Participation in Policy and Government Services 
Country	 Singapore		 Abu Dhabi
Initiative	Multi-Platform Citizen Engagement System		TAMM By You (announced Oct '25)
Description	- Integrated ecosystem of citizen participation platforms combining gamified micro-tasks, formal partnership mechanisms and deliberative engagement processes - Key platforms include (1) CrowdTaskSG (gamified micro-engagement layer) (2) Singapore Government Partnerships Office (SGPO) for citizens to submit partnership proposals across domains / areas; (3) Youth Panels for policy co-creation; (4) Forward SG (“Periodic large-scale national dialogues to refresh social compact” – entailing a policy approach that several nations and international bodies advocate to manage societal change, ensure inclusive progress and build social resilience)		- Launched as part of TAMM 4.0 enabling residents to propose ideas, vote on future platform features and shape the government services roadmap - The platform will use gamification (points system, “Hall of Fame”, leaderboard) to encourage participation in co-creating digital government experiences
Impact	- CrowdTaskSG: 50,000+ active users, with over 100 government agencies receiving over 400,000 insights from users to date; use cases include: - Gamified survey collecting youth inputs for developing a 5-year national plan for youth empowerment; ‘Great Budget Meal Hunt’ - Crowdsourced affordable meal options at HDB coffee shops; - Walking Trails@CDC: An innovative "phygital" walking programme blending outdoor activity with interactive digital challenges, where participants explore local landmarks, complete challenges and earn rewards - Forward SG: 2022-2023 exercise engaged 200,000+ Singaporeans, produced 180-page report, led to concrete policy changes (paternity leave, flexible work, housing support)		Platform just launched; unclear operational status; specific participation statistics (votes cast, features implemented) are not yet publicly available
Enablers	Tech Infrastructure: House-holds internet access, 55.1 mn smart phone subscribers, fixed broadband and strong median mobile download speak Institutional Framework: REACH (Reaching Everyone for Active Citizenry at Home) - government's policy feedback arm + SGPO as dedicated coordination body + Grassroots Organizations providing neighborhood-level engagement infrastructure Incentives: Financial incentives built into participation; integration with Redeem SG		Leverages TAMM's 3.6 million active users



- Adaptive Mechanism:** Creating structured channels for community input into policy, moving beyond ad-hoc consultation to managed platforms that can aggregate citizen insights at scale while maintaining government authority over final decisions.
- Dubai Relevance:** Dubai currently lacks a unified platform for structured community engagement. Singapore's system shows how a centralized state can maintain authority while channeling community input into policy. Abu Dhabi's “TAMM By You” signals a similar intent, extending beyond service delivery into co-design of services and features.
- Structural implications for Dubai:** Dubai's 2025 Community Participation Policy establishes the intent to channel community insights into policymaking and reinforce social cohesion. Operationalizing this will require a coordinated platform model, potentially integrated with sentiment analytics, to move from policy aspiration to functioning feedback infrastructure.

Social Adaptivity Benchmarking – Initiative Project Cards (4/5)

Cities are investing in variations of sensory and remote home care options based on contextual requirements

Component	Anticipatory Care & Vulnerability Support 	Focus Areas	Anticipatory Care 
Country	 Singapore	 Seoul	 Helsinki
Initiative	SINEW (Sensors In-home for Elder Wellbeing)	Smart Plug & AI Care Service	Remote Home Care
Description	- An anticipatory care system that uses unobtrusive in-home sensors (motion, door contact, sleep trackers) and AI to monitor seniors living alone - Does not just detect falls (reactive) but analyzes behavioral patterns to predict mild cognitive impairment and social isolation risks weeks or months before a crisis occurs	- A low-cost, high-scale operational adaptivity solution - Simple IoT "Smart Plugs" monitor electricity usage of TV / microwaves in single-person households - If usage drops to zero for a set period (e.g., 24-50 hours), an automated alert is sent to a district social worker to check on the resident	- Helsinki manages its aging population through "Remote Home Care" via tablet devices – nurses conduct scheduled video check-ins to monitor medication compliance and nutrition - System is integrated with remote monitoring peripherals (e.g., medicine dispensers) - Enables management of large caseloads by reserving time-consuming physical visits for those with high physical needs, while maintaining social contact with others, digitally
Impact	- Shifted care from "emergency response" to "preventative intervention" - Achieved >90% accuracy in detecting early cognitive decline through behavioral biomarkers alone - Allowed seniors to "age in place" independently, reducing premature institutionalization	- Drastically reduced the "discovery time" of solitary deaths (Godoksa) - Created a 24 / 7 safety net for 57,000+ households without invasive cameras or microphones 154+ emergency saves recorded in a single pilot period where lives were saved due to rapid intervention	- Over 2.5 million remote visits conducted - Studies have shown a significant decrease in emergency department visits (down ~75%) and hospitalizations for patients in the program
Enablers	Data & Sensing Ecosystems: High-fidelity sensor networks linked to the Smart Nation sensor platform Governance: Collaboration between SMU (Research), Sengkang General Hospital (Clinical) and government housing agencies Funding: Supported by the National Research Foundation and Ministry of Health to reduce long-term acute care costs	Technology Infrastructure: Simple, low-bandwidth IoT devices (LoRaWAN / WiFi) that are easy to retrofit Human Capital: The "Dolbom" (Care) SOS Centres provide the manpower to respond to alerts Institutional Incentives: City government incentivizes districts to identify "at-risk" citizens proactively	Governance: The system is run by Service Centre Helsinki, a centralized municipal unit that acts as a "Virtual Hospital." Tech Infrastructure: The "Zero-UI" passive tablet that has a distinct interface for dementia patients Robotics Integration: Enables scale via medicine-dispensing robots Data Ecosystem: Apotti, the national Electronic Health Records system



- Adaptive Mechanism:** Using ambient sensing and remote monitoring to shift elder care from reactive emergency response to preventative intervention, detecting risks such as cognitive decline, social isolation, or medical emergencies before they escalate.
- Dubai Relevance:** Dubai has piloted anticipatory care initiatives (CDA sensor pilots in 2017, DHA smart health plans in 2021), but public information on scale or continuity is limited. With the introduction of the Retirement Visa, Dubai has a greater incentive to build a citywide ambient safety system that serves both citizens and long-term residents.
- Structural implications:** Rather than deploying intrusive or high-cost monitoring systems, Dubai could prioritize opt-in, ambient models that leverage existing infrastructure (e.g., utilities, housing systems) with clear governance over alerts, escalation pathways and data use. Without defined accountability for response, sensing alone does not translate into induced adaptivity.

Social Adaptivity Benchmarking – Initiative Project Cards (5/5)

Additionally, global ‘best practices’ also reveal a focus on enabling creative social infrastructure and community ecosystems

Component	Adaptive Social Infrastructure & Community Ecosystems 	Focus Areas	Social and Innovation Anchors 
Country	 Singapore		 Helsinki
Initiative	Our Tampines Hub (OTH)		Oodi Helsinki Central Library – The "Urban Living Room"
Description	- Singapore’s first integrated community & lifestyle hub, bringing together 12 different government agencies (Sports, Library, Housing, Health, Citizen Services) under one roof - Replaces the concept of "Standalone Facilities" (e.g., a separate stadium, library and clinic) with a single "Super-Node" where diverse demographics (youth, seniors, families) are forced to collide and interact in shared common spaces - Features a 24 / 7 "Public Service Centre" where citizens can access hundreds of government services at any time		- A flagship "Non-Commercial Public Space" that redefines the library from a "Book Warehouse" to a "Makerspace for Citizenship" - Only one-third of the building stores books; the rest is dedicated to "Doing"; with installed 3D printers, laser cutters, sewing machines, gaming rooms, recording studios and VR hubs - all free to use - Acts as a "Digital-Physical Hybrid," space, providing access to high-end software / hardware to bridge the digital divide for all residents
Impact	- Footfall exceeds 20 million visits annually, turning a government facility into the town’s primary social anchor - By co-locating agencies, it reduced land usage by 30% compared to building separate facilities, while increasing service utilization within a convenient retail–oriented setting		- Busiest library in the world (on a per capita basis), with over 2 million visitors / year (in a city of ~650,000) - Democratizes access to "Future Skills" (coding, 3D modeling), turning consumers into creators - Rated as the "Most Beloved Building" in Finland, strengthening trust between the state (provider) and citizen (user)
Enablers	Governance & Decision Making: Led by the People’s Association but run by a "Place-Making" team that empowers residents to organize their own programs (e.g., Zumba, markets, volunteering) in the hub Data & Sensing: Real-time crowd analytics to manage facility usage and cooling loads; designed with an "Open feedback loop" where 15,000 residents voted on the features before construction		Regulation: Finnish law explicitly mandates libraries to promote "active citizenship and democracy," not just literacy Participatory Budgeting: The city allocated funds for residents to decide on the equipment to be housed within the library premises



- Adaptive Mechanism:** Consolidating government services, civic spaces and creative facilities into integrated community hubs that promote cross-demographic interaction and shift public infrastructure from passive service points to active social anchors.
- Dubai Relevance:** Dubai’s community infrastructure is distributed and often functionally siloed; with government services, libraries, sports facilities and coworking spaces typically split across separate sites. Libraries such as Mohammed Bin Rashid Library and Al Safa provide world-class facilities but function primarily as quiet, knowledge-consumption environments rather than maker or civic participation spaces. Converting high-traffic service nodes would shift libraries and centers from passive service points to active social and innovation anchors. A "D33 Factory" model, rebranding a public library with free-to-use 3D printers, podcast studios and laser cutters open to all residents (rather than just registered startups) could potentially democratize access to creative tools and strengthen community belonging.
- Structural implications:** Rather than expanding new standalone facilities, Dubai could increase social adaptivity by reprogramming high-traffic public assets into integrated “community super-nodes,” with explicit mandates for co-location, public access to creative tools and resident-led programming. Without this shift, community infrastructure and facilities risk remaining underleveraged in promoting social cohesion and innovation.

Environmental, Climate & Resource Adaptivity

Dubai's Environmental Agenda is driven by Clean Energy, Integrated Waste Management and Net Zero Strategies

Drawing on a review of leading global indices, this sub-section defines the core components of “environmental, climate and resource adaptivity”, assesses Dubai’s current strengths and highlights the strategic direction embedded in the city’s relevant policies. Together, these components form the analytical foundation for selecting relevant benchmark initiatives and international practices that can inform Dubai’s future adaptive pathways.

				
Component	Climate Risk Anticipation & Urban Resilience	Clean Energy Transition & Resource Efficiency	Circular Resource Systems & Waste-to-Value	Nature, Biodiversity & Blue-Green Integration
	Citywide capacity to model and prepare for heat stress, flash flooding, coastal risk and extreme rainfall	Scaling solar; reducing building / transport emissions, improving electricity–water efficiency	Reducing landfill dependency through recycling, waste-to-energy, material loops	Enhancing mangroves, shoreline resilience, shade, parks and ecological health.
Strengths	- Climate Adaptation Strategy identifies four risk pillars: heat stress, flooding / drainage, coastal risk, food-water security - Major infrastructure adaptivity investments - Strong federal early-warning systems - Dubai Urban Plan embeds climate resilience as a planning principle across land-use zones	- MBR Solar Park delivers baseload renewable capacity via 5.9 GWh thermal storage - Tasreef Deep Tunnels provide a 700% capacity surge to manage extreme 'Cloudburst' events - Integrated charging network supports projected 58,000+ EVs - World-leading grid resilience	- Dubai Waste Management Centre adds resilience by converting waste into firm energy capacity - Strong recycling systems for major material streams (construction waste, metals, tires) - National Circular Economy Policy underpins adaptive, closed-loop resource systems at the emirate level	- Expansion of reserves (Al Marmoom, Ras Al Khor) covers significant territory - Commitment to plant 100M mangroves by 2030 is active - Dubai 2040 aims to double green and recreational spaces
Focus Areas For Advancement	- Thermodynamic "Digital Twins" to model heat / wind physics - Retrofit Stress-Testing: Older building stock requires stress-testing for higher thermal loads and extreme rain events - Cross-agency climate data integration (environment, health, infrastructure) to enable faster anticipatory action	- Limited "active" participation from residents (demand response) - Building retrofit rate lags total stock growth (high growth real estate sector) - Lack of large-scale thermal batteries (e.g., aquifer or deep-water storage) to decouple cooling production from peak electrical load	- Household source separation is limited, making recycling systems more fragile - Current practice leans toward downcycling (e.g., crushed concrete) rather than higher-value urban mining and component reuse - Building materials lack digital tracking, restricting future reuse and circular construction	- Nature assets are primarily planned as landscaping or reserves, rather than as functional climate infrastructure - Need for automated / remote sensing of biodiversity health (Internet of Nature) - Green spaces are often fragmented; need for continuous cooling corridors

Climate / Environmental Adaptivity Benchmarking – Initiative Project Cards (1/5)

Adaptive Cities are pairing physical infrastructure with digital intelligence to predict and mitigate risks

Component	Climate Risk Anticipation & Urban Resilience 	Focus Areas	Pairing of physical capacity with digital intelligence for resilience 
Country	 Singapore		 Seoul
Initiative	Digital Urban Climate Twin (DUCT)		AI-Based Flood Forecasting System
Description	- A physics-based computational model that integrates environmental data (wind, sunlight, temperature) with anthropogenic data (traffic, building emissions) to simulate the city's microclimate - Unlike a standard 3D map, it calculates thermodynamic interactions		- An intelligent early-warning system that uses deep learning (long and short-term memory algorithms) to predict flood risks - It processes real-time data from rain radars, sewer water levels and river gauges to forecast inundation, up to three to six hours in advance
Impact	- Allowed Singapore to simulate "what-if" scenarios for new districts (e.g., "How will rotating this tower 45 degrees affect wind flow for the whole neighborhood?") - Identified effective cooling strategies that reduce ambient temperature by up to 0.9°C to 1.5°C in specific precincts - Shifted planning from "aesthetic greening" to "performance-based cooling"		- Reduced prediction lead time dramatically; generates forecasts at 10-minute intervals with high accuracy ($R^2 > 0.9$) - Enabled "Pre-emptive Control": The city can lower dam / reservoir levels and activate pumps before the storm peak arrives, minimizing damage during flash floods
Enablers	Data Ecosystem: Integration of high-resolution light detection and ranging data, integrated with real-time sensor feeds on city data Governance: Funded by the National Research Foundation, mandating data sharing between the Urban Redevelopment Authority and researchers (Singapore Management University / ETH Zurich) to prevent data silos Human Capital: Investment in R&D talent to build and maintain the computational engine		Technology Infrastructure: Dense IoT sensor network (water levels, flow rates) in sewers and rivers Data Ecosystem: Integration of Ministry of Environment radar data with local municipal sewer data Governance: Centralized control center (Han River Flood Control Office) authorized to trigger automated alerts



- Adaptive Mechanism:** Integrating high-fidelity digital models with physical infrastructure to enable predictive, rather than reactive, risk management. This shifts urban planning from static master plans or reactive responses to dynamic simulations that can model "what-if" scenarios before construction, thereby making build-outs more adaptive and resistant to extreme weather events.
- Dubai Relevance:** Dubai has invested heavily in large-scale engineered resilience assets (e.g., drainage, coastal protection, energy infrastructure). The city has also made notable progress in digital enabling infrastructure, including - AI-enabled operational platforms and urban data visualization tools (e.g., Dubai Here) and targeted digital-twin deployments at asset and district levels (e.g., Palm Jumeirah). However, there is no publicly available evidence of a fully institutionalized, emirate-level predictive and operational intelligence layer that continuously integrates climate sensing, simulation and decision authority into day-to-day infrastructure operations in a manner comparable to leading benchmark cities.
- Structural Implications:** Advancing toward climate adaptivity depends less on additional infrastructure investment and more on the integration of sensing, modelling and decision authority into operational workflows, enabling infrastructure to be managed dynamically rather than solely via predefined design thresholds. However, further gains in climate adaptivity are likely to depend on a UAE-level predictive climate intelligence layer that provides shared scenarios and risk insights.

Climate / Environmental Adaptivity Benchmarking – Initiative Project Cards (2/5)

Demand-side adaptivity emerges when pricing, data and incentives are aligned to influence real-time consumption behavior at scale

Component	Clean Energy Transition & Resource Efficiency 	Focus Areas	Demand side management and thermal storage 
Country	 Singapore		 Seoul
Initiative	Residential Demand Response (R-DR) Pilot & Open Electricity Market (OEM)		Eco-Mileage System
Description	The Foundation (OEM): Singapore liberalized its grid, allowing 1.4 million households to switch electricity retailers, creating a population used to managing energy costs actively The Pilot (R-DR): The Energy Market Authority launched a pilot where volunteer households with smart meters receive "Push Notifications" during grid stress events (e.g., heatwaves). If they reduce their usage during the period, they earn cash rewards which is significantly higher than the cost of power		- A voluntary incentive system where households and businesses earn "Mileage Points" if they reduce electricity, water or gas consumption by 10% or more compared to their own 2-year average - Unlike passive savings (just a lower bill), this acts as a "Cash-Back" system - with points credited twice a year - Recently evolved to link with the "Climate Card" (unlimited transit pass), allowing residents to use energy savings from their home to pay for their subway/bus rides, effectively creating a "Carbon Currency" for the city
Impact	- By aggregating thousands of households turning off appliances simultaneously, the grid gains a "Virtual Generator" of capacity without burning extra gas - Early results show participants reducing consumption by 10-40% during event windows - Accelerated the adoption of Smart Home devices (smart plugs) as residents realized they could "automate" their earnings		- Engaged over 2 million members, achieving massive aggregate energy savings - Reduced CO² emissions by approximately 2 million tons since inception (equivalent to planting 300 million trees) - Contributed to augmenting Seoul's circular economy efforts - , with accumulated points donated to "Energy Poor" families or used to buy local market vouchers, value is recycled back to the community
Enablers	Infrastructure: A rollout of Advanced Metering Infrastructure (AMI) for all households, enabling the 30-minute interval data required to verify the savings. The SP Utilities App added a "Game Mode" to the existing utility app		Institutional Incentives: Points are convertible to cash equivalents (T-money transit cards, apartment fee deductions, gift vouchers) Data Integration: The city government legally integrated billing data from KEPCO (Power), City Gas and Waterworks; residents sign up once and the data flows automatically Partnerships: Collaboration with credit card companies and retailers to redeem points



- Adaptive Mechanism:** In both Singapore and Seoul, demand-side decarbonization becomes adaptive when real-time consumption data, financial incentives and automated feedback loops are combined to influence household and business behavior during peak or stress periods, effectively turning consumers into flexible system assets.
- Dubai Relevance:** Dubai's clean-energy transition has been led by supply-side achievements. As peak demand pressures intensify, particularly during extreme heat; demand-side mechanisms that reward verified reductions in real time can complement existing infrastructure by reducing load volatility without additional capacity build-out.
- Structural Implications:** Singapore's approach relies on a liberalized retail electricity market, which is not directly transferable to Dubai. Seoul's Eco-Mileage model demonstrates that comparable behavioral outcomes can be achieved within a monopoly utility structure by overlaying incentive schemes on integrated billing data. For Dubai, this would depend on the ability to link DEWA consumption data with a citywide rewards or credits mechanism and define clear governance for verification, settlement and fraud control.

Climate / Environmental Adaptivity Benchmarking – Initiative Project Cards (3/5)

Circular Economy initiatives adopted by global adaptive cities include “Waste-to-Value” ecosystems

Component	Circular Resource Systems & Waste-to-Value 	Focus Areas	Waste prevention 
Country	 Helsinki		 Tallinn
Initiative	The Circular Construction Cluster		Circular Economy Centres & Smart Waste
Description	- A dedicated industrial and regulatory cluster focused on the "circularity of heavy materials" (concrete, bricks, glass) - Before a building is removed, a “Pre-Demolition Audit” identifies valuable components (concrete beams, windows, bricks) which are cataloged for reuse in new projects		- Tallinn is transforming traditional "waste stations" into “Circular Economy Centres” that feature repair workshops, sewing studios and "reuse rooms" where residents can swap items for free - The centers operate like modern retail spaces, using a smart booking system and license plate recognition for seamless, queue-free entry - Additionally, the city uses smart sensors in waste containers to optimize collection routes
Impact	- The flagship "Closing Loops" pilot (at Mustikkamaa) successfully built a new public service building using components salvaged from local demolition sites, proving that used materials meet safety codes - Fostered a secondary market for materials, testing digital "Material ports" that track the history and quality of used items, so architects trust them - Reduced embodied carbon in pilot projects significantly compared to new builds		- Reframes waste management from "Sanitation" to "Community Activity" - By co-locating repair cafes with recycling bins, it intercepts waste before it enters the disposal stream - Supports Tallinn's goal of 65% recycling rate by 2035 (up from ~50%), specifically targeting bulky waste (furniture) that often clogs landfills
Enablers	Incentives: City regulations requiring "circular economy criteria" in public land handover and construction tenders Human Capital: Aalto University acts as the Testing Lab, physically testing old concrete and steel to certify its safety for new buildings, removing the liability risk for private developers Data Ecosystem: "Material ports" (digital records) to track the history and quality of reused items		Strategic Rebranding: The city dissolved its "Waste Management Department" and recreated it as the "Circular Economy Department," legally changing the mandate from handling waste to preventing it Partnerships: Collaboration with NGOs to manage reuse operations Technology Infrastructure: IoT sensors in bins communicating via cellular network



- Adaptive Mechanism:** Moving beyond recycling into industrial ecosystems that treat waste as an economic resource; shifting from "demolish and crush" to "deconstruct and reuse" for construction materials; and from "sanitation" to "community activity" for household waste.
- Dubai Relevance:** Dubai’s scale of construction, refurbishment and consumption creates substantial latent material value. As development in the Emirate further intensifies, approaches that prioritize deconstruction, certified reuse and community engagement can complement existing waste-to-energy capacity by reducing the loss of high-value material.
- Structural Implications:** Applying these models would require:
 - Regulatory recognition and reward of reuse and deconstruction as compliant alternatives to demolition and disposal
 - Trusted certification mechanisms (digital or institutional) to reduce liability and quality concerns around reused materials
 - Physical or digital marketplaces that connect supply (demolition, refurbishment, household turnover) with demand (developers, SMEs, residents), rather than treating circularity solely as a waste-management function

Climate / Environmental Adaptivity Benchmarking – Initiative Project Cards (4/5)

Integrated waste-water-energy systems enable resource adaptivity by maximizing recovery value across traditionally siloed infrastructure

Component	Circular Resource Systems & Waste-to-Value 	Focus Areas	Waste-energy nexus 
Country	 Singapore		 Denmark
Initiative	Tuas Nexus		Billund BioRefinery
Description	- The world’s largest integrated facility co-locating a Water Reclamation Plant (TWRP) and an Integrated Waste Management Facility (IWMF) on a single site (saving 2.6 hectares of land) - Waste heat (steam) from the IWMF is used to dry sludge from the TWRP, while food waste slurry is piped to the water plant for co-digestion, maximizing biogas production		- A municipal wastewater treatment plant retrofitted into a "Bio-Refinery" that processes both sewage sludge and organic waste (food waste from households / industry) - Uses the Exelys™ technology to "pressure cook" the mixed sludge before digestion, making the material easier to break down biologically and drastically increasing gas yields
Impact	- The facility is 100% energy self-sufficient, generating enough power from trash to run both plants, with excess electricity exported to the national grid - Saves 200,000 tons of CO2 annually - Maximizes land use and resource recovery through integrated design		- Achieves a massive increase in biogas production, producing 160% more energy than the plant consumes, turning the facility into a net exporter of electricity and heat - Reduces the amount of sludge by 30%, which is then converted into high-quality fertilizer (struvite) for local farms
Enablers	Governance: Formal Joint Venture between PUB (Water Agency) and NEA (Environment Agency) to break down traditional agency silos and jointly tender the infrastructure		Technological Upgrade: Strategic investment in thermal hydrolysis as the core technology to unlock the energy potential of food waste and sludge co-digestion Financial Alignment: Integration with local utilities and farmers ensures a market for the energy and fertilizer outputs



- Adaptive Mechanism:** Across benchmark cities, adaptivity emerges when wastewater, organic waste and energy systems are planned and operated as a single resource loop. Co-location and shared process design allow waste streams to be dynamically redirected toward energy recovery, rather than treated as fixed disposal liabilities.
- Dubai Relevance:** Dubai has invested in large-scale “waste-to-energy” infrastructure to reduce landfill dependence and generate power. However, unlike integrated nexus models (e.g., Singapore’s Tuas Nexus), waste, water and energy systems remain planned and operated largely within separate institutional mandates, limiting full system-level optimization.
- Structural Implications:** Applying this approach would require:
 - Joint planning and performance accountability across water, waste and energy authorities
 - Outcome metrics that prioritize net energy and resource recovery, not only treatment volumes
 - Early integration of waste-energy considerations into greenfield and major brownfield expansions, rather than retrofitting after asset build-out and delivery

Climate / Environmental Adaptivity Benchmarking – Initiative Project Cards (5/5)

Cities adopt nature-based solutions that reflect their ecology and urban form

Component	Nature, Biodiversity & Blue-Green Integration 		Focus Areas	Natural defense and nature-based solutions 
Country	 ABU DHABI	 SEOUL		 TALLINN
Initiative	Blue Carbon Drone Restoration	"Baram-gil Sup" (Wind Path Forests)		The Pollinator Highway (Putukaväil)
Description	- The use of custom-engineered heavy-lift drones to plant mangrove seeds in difficult-to-access coastal mudflats - The drones use imaging to identify optimal soil areas and drop seeds at specific velocities	- A strategic forestry project that creates "Wind Paths" to channel fresh, cool air from the surrounding mountains (Gwanaksan / Bukhansan) into the city center - It uses three types of forests: Wind-generating (mountain), Connecting (riverside / roadside) and Spreading (urban parks)		- A 13.5km linear park built along a former high-voltage power line corridor and railway embankment - It connects six districts and is specifically designed to support pollinators (bees, butterflies) while providing a green transit route for people
Impact	- Scaled planting to millions of seeds with a success rate exceeding 35% (high for automated planting) - Reduced planting costs and labor intensity significantly compared to manual methods - Established a scalable "Bio-Shield" against coastal erosion	Cooling: Studies show the forests reduce average temperatures in downtown Seoul by 3°C to 7°C during summer Air Quality: Specifically designed to flush out particulate matter (PM2.5) and stagnant heat from the urban basin		Biodiversity: Created a continuous habitat supporting 42 butterfly species and 22 bumblebee species in the heart of the city Urban Repair: Converted a "non-place" (industrial utility corridor) into a vibrant public space without requiring building demolition
Enablers	Technology: Partnership with drone specialists (Distant Imagery) and energy companies (ENGIE) Funding: Corporate CSR funding channeled into quantifiable "Blue Carbon" credits Governance: Environment Agency – Abu Dhabi provided scientific mapping of planting zones	Data & Sensing: Detailed wind path analysis using German wind-mapping technology (KALM) to identify where to plant trees to maximize airflow Funding: Joint funding (17 billion KRW) between the Korea Forest Service and Seoul Metropolitan Government		Governance: Cross-departmental collaboration (Urban Planning, Environment and Transport) Citizen Engagement: Extensive co-creation workshops with residents to design the space



- Adaptive Mechanism:** Global benchmark cities demonstrate that nature-based solutions are most adaptive when they are designed and governed as functional urban infrastructure, not solely as conservation or beautification initiatives. Cooling, airflow, erosion control and biodiversity connectivity are treated as explicit performance objectives.
- Dubai Relevance:** Dubai already undertakes large-scale greening, mangrove expansion and reserve protection as part of its environmental agenda. As climate pressures intensify, the adaptive opportunity lies in more explicitly linking these interventions to defined urban performance roles (e.g., passive cooling corridors, coastal bio-shields, biodiversity connectivity across infrastructure).
- Structural Implications:** Moving toward adaptive use of nature-based solutions requires a more explicit linkage between planning intent, infrastructure performance and long-term management responsibilities. This does not imply the need for new strategies, but a sharper articulation of the functional role of nature within existing planning, zoning and infrastructure decision-making frameworks, supported by data and cross-agency coordination.

Urban Infrastructure & Mobility Adaptivity

Dubai's Urban Agenda is driven by Urban Plan 2040 and Autonomous Transportation Strategy

Drawing on a review of leading global indices, this sub-section defines the core components of “urban infrastructure and mobility-based adaptivity”, assesses Dubai’s current strengths and highlights the strategic direction embedded in the city’s existing policies. Together, these components form the analytical foundation for selecting relevant benchmark initiatives and international practices to inform Dubai’s future adaptive pathways.

				
Component	Mobility & Network Adaptivity Ability of the transport system to sense, manage and adapt to congestion, modal shifts, climate events and population growth.	Infrastructure Resilience & Continuity of Essential Services Capacity of utilities, cooling, drainage, and emergency systems to withstand disruptions and maintain service continuity.	Sensorised Operations & Intelligent City Management Real-time monitoring, analytics and predictive capabilities that allow proactive, integrated management of the urban system.	Spatial Flexibility & Public Realm Adaptivity Ability of the built environment to evolve walkable, accessible and responsive to population and lifestyle changes.
Strengths	- Highly integrated RTA traffic and transport management system - Expanding multi-modal network (metro, tram, marine, first / last mile) - Autonomous Transport Strategy targets 25% autonomous trips by 2030 - Rapid growth of EV charging, micromobility regulation and smart parking	- DEWA recognized globally for reliability, grid redundancy and service continuity - Major stormwater upgrades towards draining security (e.g., Deep Tunnel Stormwater System) - Established large-scale district cooling networks, with the ability to shift peak electrical load using thermal energy storage tanks for daily resilience - Strong emergency and civil defense response capabilities	- Extensive IoT and smart-system integration via Digital Dubai - RTA uses real-time sensors and AI for adaptive traffic control - DEWA smart grid enables predictive maintenance and high uptime - Citywide dashboards provide operational situational awareness	- Dubai 2040 embeds 15-minute city, green corridors, and walkability principles - Strong delivery capacity for major public-realm upgrades - High-quality shaded and climate-adaptive environments in newer districts
Focus Areas For Advancement	- Heat-resilient first / last mile: expand shaded “Walk2Ride” corridors - Dynamic multi-modal optimization - Embedded climate stress tests for mobility infrastructure (extreme rainfall, heat)	- Stronger utility risk forecasting using integrated climate and operations data - Extension of utility-level predictive operations into cross-system anticipatory coordination - Enhance district cooling efficiency and peak-resilience upgrades - Continued and targeted drainage upgrades in older districts	- Unified resilience dashboard integrating environment, mobility, utilities, emergency data - Broader predictive-maintenance coverage for ageing infrastructure - Real-time public alerts on operational impact, for extreme weather and disruptions - Expanded IoT sensing in older neighborhoods (heat, air quality, flood)	- Review tendering and regulatory pathways to test emerging urban-tech solutions “Active” climate-controlled corridors in key business areas (as shading remains inadequate in extreme heat)

Urban Infrastructure & Mobility Benchmarking – Initiative Project Cards (1/5)

Adaptive mobility systems treat environmental comfort as a core transport performance requirement...

Component	Mobility & Network Adaptivity 	Focus Areas	Last-Mile Connectivity 
Country	 Singapore		 Seoul
Initiative	Walk2Ride Programme & Integrated Climate-Controlled Links		Cheonggyecheon Stream Restoration & Pedestrianization
Description	- Uniform, oppressive tropical heat and high humidity discourages outdoor travel - The Walk2Ride Programme mandates a continuous, sheltered network (within 400m of transit stations) connecting residences and amenities - This network anchors into “Integrated Transport Hubs”: large, fully air-conditioned complexes that blend the Bus Interchange, MRT station and adjacent commercial buildings		- Extreme cold in winter and heat / monsoon flooding in summer, is exacerbated by a dense urban core (Urban Heat Island effect) - The Cheonggyecheon Stream Restoration & Pedestrianization Project dismantled a 5.6km elevated highway to restore the original stream, creating a pedestrian-only corridor with extensive tree cover and cooling shade. The open waterway now functions as a critical flood-control channel that manages monsoon runoff
Impact	- Provides a comfortable, air-conditioned link, removing the primary barrier to walking and ensuring the last mile is viable 365 days a year - Achieved sheltered access within 400m of all key transit nodes, making walking the default option for short distances in core areas - Reduces commuter friction, making multi-modal public transit a more appealing choice than individual car use		- Successfully reduced ambient street temperatures by 3°C to 6°C compared to nearby parallel roads - Increased local biodiversity by over 600% and reduced small-particulate air pollution (PM2.5) by 35%
Enablers	Regulation: Land Transport Authority mandates sheltered access as a requirement for all new major transit hubs Economic Viability: High capital investment is justified by the economic value of the high-density commercial assets (e.g., shopping centers, financial towers) the links serve		Political Will: The success hinged on political leadership prioritizing quality of life over traffic convenience (requiring multi-modal transit expansion to handle displaced cars) Engineering: Advanced hydraulic engineering was used to reconstruct the stream bed to withstand a 200-year flood interval



- Adaptive Mechanism:** Benchmark cities improve “first to last mile” performance by explicitly addressing thermal comfort and environmental exposure as functional constraints on public transport usage. This is achieved either through regulated, climate-controlled pedestrian links at major hubs (Singapore) or through district-scale public-realm adaptations - such as shade, de-paving and water-based cooling, where enclosure is impractical (Seoul).
- Dubai Relevance:** Dubai has piloted elements of both approaches in selected developments. However, neither model has been deployed at a citywide scale, reflecting the cost, spatial and feasibility trade-offs between comprehensive climate-controlled infrastructure and lower-intensity public-realm interventions across a dispersed urban form.
- Structural Implications:** Rather than pursuing uniform solutions, Dubai’s adaptive pathway potentially lies in differentiated, district-specific “first to last mile” strategies, embedded into transit-oriented development standards and major mobility nodes, while recognizing that full year-round pedestrian comfort may be achievable only in targeted locations.

Urban Infrastructure & Mobility Benchmarking – Initiative Project Cards (2/5)

Network synchronization and multimodal integration represent different approaches to making public transport competitive with car ownership

Component	Mobility & Network Adaptivity 	Focus Areas	Multimodal Transport 
Country	 Zurich		 Helsinki
Initiative	The "Taktfahrplan"	Whim App (Discontinued) & The Lessons of Mobility as a Service (MaaS)	
Description	- A fixed-interval, rhythmic timetable where all public transport services (trains, trams, buses) operate at standardized, regular intervals (e.g., every 15, 30, or 60 minutes) - The entire network is designed so that multiple lines converge at key interchange nodes at the same moment, allowing passengers to step off a train and immediately onto a waiting bus or tram with minimal layover time	- Aggregated planning, booking and payment for all transport modes (public transit, taxi, rental car, bike-share, micromobility) into a single application and sold it as a subscription service - Focused on selling the journey (A to B) rather than the mode (bus ticket + taxi fare), aiming to provide convenience that rivals and eventually replaces, private car ownership	
Impact	- Achieved the highest per-capita public transport usage in the world, proving that reliability and synchronization drive modal shift more effectively than speed alone - Allows for easy network expansion (adding a new feeder bus line doesn't disrupt the core timetable) and simpler communication to users	- Successfully validated consumer demand for "mobility bundles" and shifted users away from private cars - The B2C model filed for bankruptcy in 2024 because it could not negotiate low enough wholesale rates with private transport operators to make the subscription profitable	
Enablers	Governance: The Zurich Transport Network coordinates schedules across 37 transport companies through service contracts and a legislated timetabling ordinance Infrastructure Design: Prioritizes smooth transfer times (usually 3-5 minutes) over maximum line speed, treating the transfer itself as part of the primary journey	Legislation and Standardized Data: The Act on Transport Services mandated open APIs, forcing all transport operators to share ticketing and routing data with third-party aggregators Technology: Single payment channel	

- 
 - Adaptive Mechanism:** Multimodal adaptivity is achieved when network coordination, scheduling logic and fare integration are coordinated at the network level - so transfers across modes are predictable and low-friction. Zurich operationalizes this through synchronized timetables and timed transfers embedded in regulation and planning. Helsinki demonstrated that app-led Mobility-as-a-Service (MaaS) without structural alignment of operator incentives is insufficient.
 - Dubai Relevance:** Dubai has adopted a centralized, app-based aggregation approach through S'hail, which integrates journey planning and payment across Metro, bus, taxi and selected ride-hailing services. This has improved visibility and convenience but primarily supports pay-per-ride, mode-specific travel. The key learning from the assessed benchmarks is therefore not about introducing new modal access platforms, but about whether multimodal travel is supported through coordinated timetables, transfer conditions and fare logic that reduce waiting time and uncertainty at interchanges.
 - Structural Implications:** Dubai does not require institutional reform to strengthen multimodal adaptivity, as the RTA already plans, regulates and operates most core transport services. The structural shift lies in explicitly embedding transfer reliability, interchange performance and network-level KPIs into planning, regulation and fare-setting processes, enabling multimodal journeys to be optimized as end-to-end trips rather than as a sequence of independent services.

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Urban Infrastructure & Mobility Benchmarking – Initiative Project Cards (4/5)

Global adaptive cities also offer contextual models for building integrated, sensorized urban operations

Component	Sensorised Operations and Intelligence City Management 	Focus Areas	Smart city systems 
Country	 Singapore		 Seoul
Initiative	Smart Nation Sensor Platform (SNSP)		Smart Seoul Platform
Description	- The SNSP is a "Whole-of-Government" integrated system. It connects >110k sensors to a central data platform to automate city management, allowing the government to address issues (such as water leaks or traffic jams) proactively before they disrupt citizens <i>Core Philosophy:</i> Efficiency, predictive maintenance and seamless government operations		- Seoul leverages data to empower citizens and businesses. While the Digital Mayor's Office visualizes the data, the Smart City Integrated Platform is the engine that physically connects the police, fire and municipal CCTV networks for immediate crisis response <i>Core Philosophy:</i> Transparency, citizen participation and public-private value creation
Layers	Infrastructure: AG (Aggregation Gateway) box attached to streetlights; provides power and connectivity to sensors from any agency, eliminating the need to dig separate trenches IoT Platform: DECADA, the central operating system that ingests and standardizes data from diverse sensors; the system uses AI to detect anomalies Exchange: APEX, the API that enables agencies to share real-time data streams securely without complex bilateral agreements; APEX is a proprietary, government-built secure gateway Verification: National digital identity framework (Singpass / Corp) validates the identity of citizens reporting issues and allows for location-specific notifications Action: The front-end applications that route verified alerts to the correct agency automatically, or work orders to contractors, or deliver targeted notifications to citizens (e.g., flood warnings)		Infrastructure: S-Pole is a custom-designed pole that physically integrates traffic lights, streetlamps, CCTVs and S-DoT sensors into one unit IoT Platform: Smart City Integration Platform feeds from the S-Poles directly to the relevant agencies on crime, disasters, traffic and other emergencies Exchange: Open Data Plaza releases non-sensitive data from the S-Poles (e.g., footfall, weather) to the public for innovation; 'NGSI-LD' allows any external developer or vendor to easily "plug in" to city data Verification: Uses blockchain to verify residency without revealing personal data Action: Citizens vote on public issues based on data; Private system operators can receive insights on gaps, build relevant solutions and co-create with the government

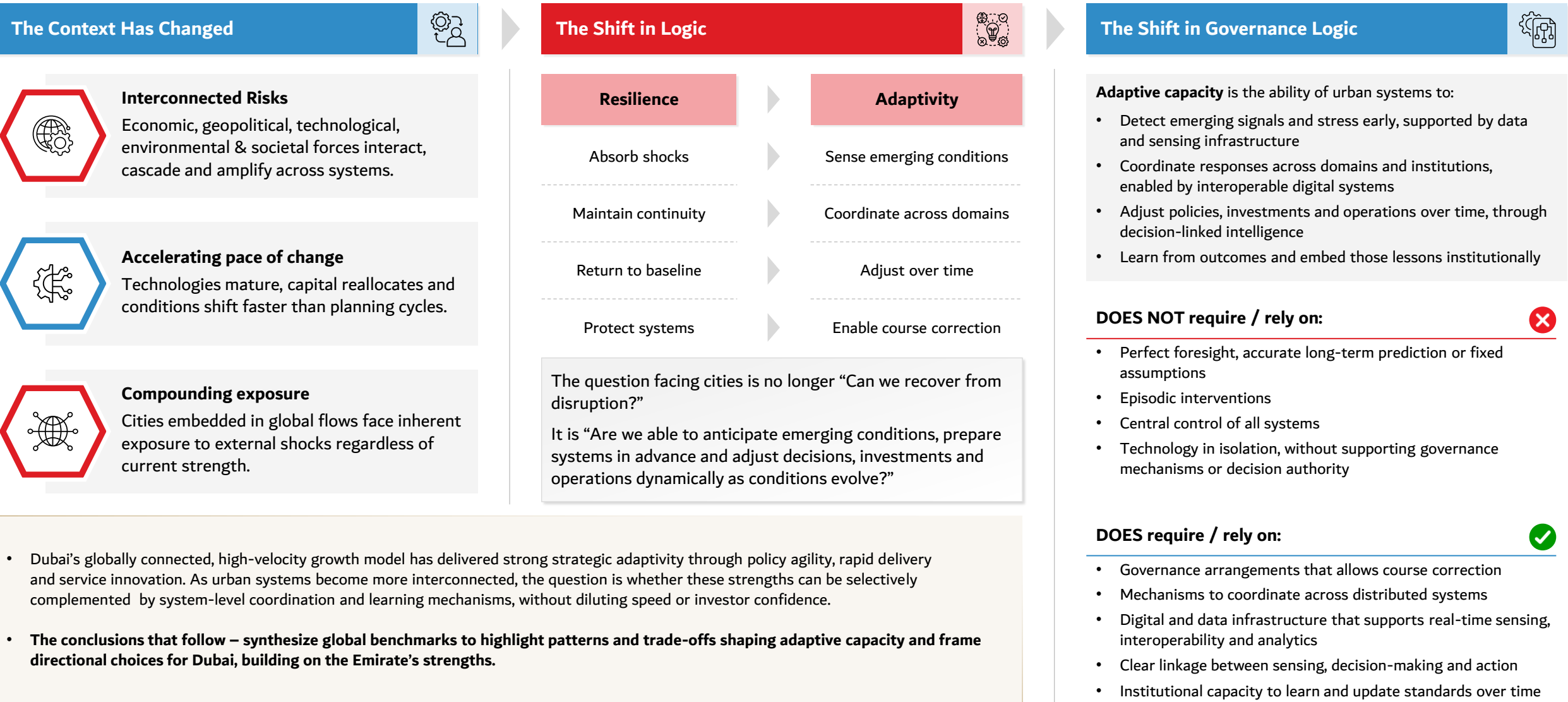
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- Adaptive Mechanism:** Across benchmark cities, urban operations become adaptive when sensing, data standardization and response workflows are institutionally linked, allowing signals from the physical city systems to trigger coordinated operational actions rather than remain informational dashboards. Singapore emphasizes “Whole-of-Government” operational integration; whereas Seoul demonstrates the value of open, shared data ecosystems, in which citywide sensor outputs are made available to public agencies and innovators, thus enabling high-value mobility, safety and civic applications.
 - Dubai Relevance:** Dubai’s model remains entity-owned and entity-controlled, with sensors, operational platforms and real-time data flows being fragmented across multiple agencies such as the RTA, DEWA, Dubai Police, Dubai Municipality and free zones. This distributed structure means that the Dubai Pulse platform currently functions as a data aggregation layer, rather than a unified operational platform, as it lacks consistent real-time feeds, multi-data-source integrations via APIs and shared action protocols. The potential benefit of an integrated sensor-based intelligence system is significant: faster emergency response, reduced operational duplication, dramatically improved predictive maintenance and a foundation for anticipatory, AI-enabled urban management – aligned with Dubai’s smart mobility ambitions.
 - Structural implications for Dubai:** Moving toward integrated, sensor-driven urban operations depends on common data standards, interoperable APIs and clearly defined response authority across entities. Without these structural linkages, sensor platforms function primarily as aggregation layers rather than operational systems capable of anticipatory city management.

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02	Part 1: Key Learnings from Structural Models – Singapore and Abu Dhabi
03	Part 2: Comparative Benchmarking of Domain-Specific Initiatives and Enablers
04	Conclusion and Directional Recommendations for Dubai



Conclusion

Adaptivity is becoming a critical requirement for global cities



Conclusion

Benchmarks indicate that adaptive capacity is primarily shaped by structural configurations, rather than individual, standalone initiatives

Across global benchmarks, adaptive outcomes are evidently shaped by structural conditions (inherited or deliberately designed) rather than by individual technologies, platforms or programmers.



Key Findings



Comparable initiatives produce different adaptive outcomes depending on structural context

Models for digital twins, life-journey platforms, or economic foresight units function differently depending on how sensing, coordination, decision-making and feedback are institutionally linked. Where such linkages are weak, initiatives remain isolated improvements; where they are embedded, initiatives contribute to cross-domain or system-wide adaptivity. Dubai has successfully deployed advanced projects, systems and programmes across multiple domains. The extent to which these initiatives reinforce cross-domain or system-wide adaptivity will depend on how effectively they are institutionally connected across entities and domains.



Adaptive outcomes reflect path-dependent structural trade-offs that are not universally transferable

The adaptive characteristics of highly centralized models (Singapore) or state-capital-led models (Abu Dhabi) reflect specific political, fiscal and institutional trajectories that cannot be directly replicated within Dubai’s federated, market-oriented framework.



Adaptivity operates at multiple levels, each with distinct structural requirements

Initiative-level and domain-specific adaptivity can be achieved without system-level adaptivity. Dubai’s strongest adaptive performance appears at the initiative and domain levels where mandates are clear, delivery authority is concentrated, and feedback loops are operational.



Adaptive capacity may be selectively built and incrementally built

Benchmark evidence shows that adaptivity does not emerge uniformly across cities, but varies by domain according to risk exposure, development models and governance arrangements. In Dubai as well, this indicates that adaptive capacity has developed unevenly across sectors, strengthening first where these structural conditions are most closely aligned.

Conclusion

These structural configurations are underpinned by cross-domain enablers and mechanisms

Technology Infrastructure

- Digital backbone including connectivity networks, computing platforms, cybersecurity systems and service delivery channels that enable adaptive operations

Enabler / Mechanism	Benchmark Reference
Backend Data Interoperability & Shared APIs	Singapore’s APEX, Tallinn’s X-Road, Seoul ‘s NGSI-LD
Integrated Service Delivery Platforms	Singapore’s LifeSG, Abu Dhabi ‘sTAMM 4.0

Consideration: Technology infrastructure enables adaptive capacity when backend interoperability supports cross-entity coordination and real-time decision-making.

Governance, Decision Making & Institutional Incentives

- Institutions, policies, regulations, participatory processes that authorize, coordinate and legitimize cross-domain adaptive actions
- Budget mechanisms, performance metrics and reward structures that align departmental goals with system-wide outcomes

Enabler / Mechanism	Benchmark Reference
Coordination Authority & Mandates	Singapore’s SNDGG, Abu Dhabi’s DGE
Community Participation in Policy-Making	Singapore’s CrowdTaskSG / Forward SG
Planning Frameworks with Adaptive Requirement	Kalasatama’s (Helsinki) agile piloting
Cross-Entity Performance Alignment	Finland’s outcome-based budgeting and performance frameworks; Singapore’s “Whole-of-Government” performance and accountability model

Consideration: Governance arrangements determine whether adaptive responses can be authorized, funded & executed across entity boundaries.

Data & Sensing Ecosystems

- Sensors, data standards, interoperability protocols, analytics platforms and feedback mechanisms that generate situational awareness and actionable intelligence

Enabler / Mechanism	Benchmark Reference
Economic Foresight and System Sensing	Singapore Pulse of Economy; RAHS; Centre for Strategic Futures
Predictive Climate & Infra Operations	Singapore DUCT, Seoul AI Flood Forecasting
Integrated Urban Sensing and Data Platforms	Singapore SNSP, Seoul Smart City Platform
Real-Time Operational Dashboards	Seoul TOPIS

Consideration: Data and sensing systems support adaptivity when signals are shared, analyzed jointly and linked to decision authority.

Human Capital & Innovation Capacity

- Skilled workforce and innovation ecosystems (research institutions, startups, training programs) that generate knowledge and develop adaptive solutions

Enabler / Mechanism	Benchmark Reference
Economic-Skills-Firm Linkage	Singapore ITM-JTM-OTR, SkillsFuture, WSG
SME Capability Upgrading Ecosystems	Singapore Queen Bee, Seoul Win-Win Smart Factory
Systems and Data Talent Embedded in Government	Singapore GovTech; Estonia digital units

Consideration: Human capital enables adaptivity by supporting experimentation, learning and institutionalization of new approaches.

Across benchmarks, adaptive capacity is strongest where data, authority, coordination, incentives and learning are aligned..

*Benchmark cities exhibit varying degrees of cross-enabler alignment.

Conclusion

What global indices indicate about structural translation and compounding of outcomes

Index		What the Index Measures	Dubai Ranking
	IMD Smart City	Resident perception surveys (≈120 respondents per city) combined with contextual indicators assessing how effectively technology supports urban outcomes across mobility, governance, health and safety, activities and opportunities.	4
	IESE Cities in Motion	~100 indicators across 9 dimensions: Economy, human capital, social cohesion, environment, governance, urban planning, international outreach, technology and mobility.	81 (2 nd regional)
	Kearney Global Cities Resilience	Global city performance across business activity, human capital, information exchange, cultural experience and political engagement.	4
	Mercer Quality of Living	~40 factors assessing expatriate quality of living, including social and political environment, economic conditions, healthcare, education, housing, infrastructure, recreation and natural environment	83
	GSER	Startup ecosystem maturity measured through ecosystem value, funding, exits, talent depth and knowledge spillovers	Not in top 40, (2 nd in MENA)
	Arcadis Sustainable Cities	Sustainability outcomes across Planet, People and Profit dimensions (with recent editions adding a transition / progress lens)	Not ranked in top 100

Dubai – Overview of Index Performance*

- Dubai demonstrates strong performance on indices focused on development and enablement, with excellence demonstrated on indicators related to technology readiness, digital innovation, safety and security, infrastructure, institutional capacity, start-up funding, international profile and attracting international talent. Notably, Dubai’s social cohesion and community participation policy was explicitly recognized in global resilience assessments.
- In contrast, Dubai’s performance is less pronounced on indices that place greater weight on cumulative outcomes, including indicators related to environmental sustainability, human capital depth, knowledge (innovation & research output), economic ecosystem maturity, housing affordability and infrastructure intensity.
- Overall, index results indicate that Dubai performs consistently well where indices measure intent (through strategies and policies), activity, investment and implementation capacity. On a relative basis, areas of improvement emerge on specific indices that assess how outcomes accumulate and stabilize over longer time horizons. This points to an opportunity to strengthen how Dubai’s deployed capabilities and high levels of activity can be translated into durable, system-wide outcomes over time.

*Most indices do not disclose full indicator weightings or aggregation methods; results should be interpreted as directional signals rather than precise measurements of domain-level performance

Directional Recommendations

Structural considerations for strengthening adaptive capacity (1/3)

Taken together, the indices and benchmarks indicate that Dubai’s challenge lies less in initiating action and delivering new projects and more in how outcomes accumulate across the system over time.

The emergent imperative is to deepen coordination selectively in priority domains, while preserving the speed, commercial discipline and sector-led excellence that define Dubai’s competitive positioning.

The recommendations that follow therefore focus on governance and operating-model interventions that strengthen coordination, learning and system-level decision-making across existing efforts.



Overall Direction: *Preserve Dubai’s delivery strengths while selectively deepening coordination where system value is highest.

Preserve	Deepen Selectively (Where System Value is Highest)
Execution speed and visible delivery	Backend interoperability
Customer-centric service innovation	Cross-entity data sharing in priority domains
Private-sector leverage and FDI attraction	Planning frameworks that embed learning and iteration
Regulatory agility and experimentation	Demand-side activation (resident / citizen participation)
Competitive dynamics driving improvement	Coordination authority in domains with highest system risk

This:

- Builds on already emerging pilots towards deeper integration, such as NABIDH (health data), Unified License, work bundle and digital twin partnerships
- Avoids radical changes in existing governance structures
- Institutionally feasible within Dubai's federated model
- Generates incremental gains that compound over time

*This direction frames how the recommendations that follow should be interpreted.

Directional Recommendations

Structural considerations for strengthening adaptive capacity (2/3)



Directional Recommendations

1



Mandate system-level learning from major programmes and districts

- Major programmes and selected districts should be designed to generate ongoing, reusable system-level insights that inform future standards, funding rules and planning frameworks.
- This requires:
 - Standardized processes to capture: Operational performance, behavioral response, cost and lifecycle implications, coordination and implementation risks
 - Clear pathways for translating insights into: Updated standards and design guidelines, funding and investment criteria, planning and regulatory frameworks

2



Define high-interdependence areas and associated cross-domain decision protocols

- Areas of high interdependence should be explicitly identified and designated where cross-sector interactions create material system-level risk that cannot be managed through existing bilateral partnerships or entity-level coordination.
- For these defined cases, cross-domain decision alignment may be formalized through:
 - Clear criteria for designation, specifying when an issue qualifies as system-level (e.g., multi-entity dependency, cascading impacts, or irreversible consequences)
 - Defined decision rights and escalation protocols for resolving specified trade-offs once a case meets these criteria
 - Shared outcome indicators that reflect system-level performance rather than entity-level performance

3



Pre-define system-level triggers based on recurring operational indicators

- Pre-define a small set of recurring operational indicators that, when observed across multiple entities or locations over time, automatically trigger system-level planning, investment, or regulatory review.
- This requires:
 - Agreement on a limited set of system-level indicators, such as sustained congestion across corridors, repeated service backlogs in certain domains, persistent capacity shortfalls or recurring demand shifts that undermine planning assumptions
 - Pre-defined links between indicators and decisions, specifying which planning standards, investment sequencing, funding rules, or regulatory settings must be reviewed when each indicator is triggered
 - Clear escalation rules, defining when responsibility moves from operational management to cross-domain decision forums

Directional Recommendations

Structural considerations for strengthening adaptive capacity (3/3)



Directional Recommendations

4



Formally designate learning districts to test and scale priority policy objectives

- Formally designate a limited number of districts as learning districts, where experimentation, structured community participation and evidence-based iteration are explicitly required in order to test, refine and scale priority policy objectives (e.g., community participation in design, service integration, sustainability or new planning standards) before wider system adoption.
- This requires:
 - Explicit designation of learning districts aligned to specific policy priorities or system outcomes the city seeks to scale
 - Clear specification of what differs in these districts including: Which planning, regulatory or operational standards may vary within defined guardrails and what data and resident / user feedback must be collected
 - Defined decision consequences specifying: How evidence from learning districts is used to scale, modify or discontinue approaches and how it informs updates to planning standards, funding rules and regulatory frameworks
 - Structured participation requirements embedding resident and user feedback as a formal input to learning, consistent with Dubai's community participation objectives and supported by clear guidance and proportionate incentives where delivery is led by private developers

5



Differentiate investment governance for adaptive and experimental initiatives

- Explicitly differentiate how adaptive and experimental initiatives are approved, funded and evaluated, recognizing that initiatives designed to test, learn or manage uncertainty require different governance from standard delivery projects.
- This requires:
 - Clear classification of adaptive initiatives: Defining which initiatives qualify as adaptive or experimental, based on uncertainty, system impact or learning intent
 - Adjusted approval and evaluation criteria: Allowing learning outcomes, risk reduction and option value to be recognized alongside delivery milestones and financial performance
 - Protected funding and review pathways: Enabling adaptive initiatives to proceed without being prematurely constrained by traditional business-case assumptions, while retaining clear stop, scale or redesign decisions

Directional Recommendations

Domain Specific Directions (1/7)

Approximately 21 themes were identified across five adaptivity domains. The initiatives that follow represent a selection of directional recommendations that present actionable recommendations for Dubai. Final prioritisation should reflect Dubai's strategic objectives, implementation feasibility and resource constraints. As benchmark cities do not demonstrate uniform adaptive capacity across all domains, selectivity is both necessary and appropriate – the strategic choice lies in where Dubai chooses to focus and lead.



Economic Adaptivity

Future economy preparedness through sector transformation planning

- Develop sector transformation roadmaps that anticipate how priority sectors will evolve and translate this insight into coordinated action across firm and SME capabilities, workforce roles and skills, technology adoption and supporting policy and investment sequencing
- These roadmaps should move beyond high-level strategy to specify:
 - How value chains are expected to change;
 - Which activities will grow, automate, relocate, or decline;
 - What capabilities firms, particularly SMEs, will require to remain competitive;
 - How workforce roles and skills will shift and
 - What are the enabling actions the public sector must sequence in relation to skills development, incentives, infrastructure and regulation
- The objective is to prepare firms, workers and institutions for sectoral transition through plausible pathways that reduce adjustment costs and time lags as sectors evolve. This approach also informs Dubai's future global talent requirements, supporting targeted attraction and retention through immigration and complementary mechanisms
- **Benchmark Reference:** Singapore ITMs, OTRs, JTM, linking to SME development and skill development programmes
- **Outcome Dimensions Supported:** Economic adaptability; SME capability and productivity; Human capital relevance; Ecosystem maturity



Economic Adaptivity

Ecosystem-based SME capability transfer

- Establish structured mechanisms that enable anchor firms to transfer operational, technological and managerial capabilities to SMEs through commercial relationships. This accelerates capability diffusion across value chains, particularly for SMEs that face constraints in upgrading independently.
- Benchmark Reference:** Singapore Queen Bee, Seoul Win-Win Smart Factory
- Outcome Dimensions Supported:** SME productivity and competitiveness; Ecosystem depth; Supply-chain resilience; Inclusive economic growth

Foresight-driven resource allocation through institutionalized horizon scanning

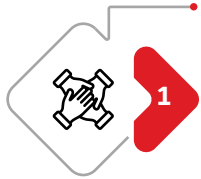
- Establish a formal mechanism that links early-warning signals and scenario planning to budget and policy decisions, ensuring foresight directly informs resource allocation rather than remaining advisory in nature. This would involve institutionalizing horizon scanning within central government, building cross-agency situational awareness for emerging shocks and creating formal linkages between foresight outputs and budgetary and policy cycles.
- The aim is to move from reactive crisis response to anticipatory resource positioning, reducing lag between signal detection and government action.
- Benchmark Reference:** Singapore CSF within PMO Strategy Group, RAHS, Pulse of the Economy
- Outcome Dimensions Supported:** Governance quality; Economic resilience; Strategic planning maturity

Innovation pipelines and commercialization pathways

- Strengthen end-to-end innovation pipelines that connect research, start-ups, corporates and public demand, reducing drop-off between experimentation, adoption and scale. This includes clarifying pathways through which pilots, proofs of concept and early-stage ventures progress to large-scale deployment, sector-wide adoption, productivity gains or commercial exits. The emphasis is on complementing strong early-stage activity with mechanisms that enable first adoption, scale-up and diffusion across firms and sectors (rather than expanding experimentation on a standalone basis).
- Benchmark Reference:** Singapore – use of testbeds, public procurement and sector platforms to translate innovation into scaled adoption and productivity outcomes
- Outcome Dimensions Supported:** Ecosystem maturity; Innovation diffusion; Productivity growth; Commercialization outcomes

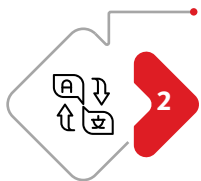


Social Adaptivity



Community participation as a continuous feedback mechanism

- Build on the Community Participation Policy by using continuous feedback channels, rather than one-off consultations. The objective would be to identify emerging needs, priorities and service gaps, ensuring community signals consistently inform service and intervention refinement over time.
 - Structured approaches to collecting, aggregating and interpreting community input at scale, with clear links between participation outputs and defined policy and service-design decisions
 - Integrating resident and user feedback into infrastructure and service design, particularly in areas undergoing rapid change or experimentation
- **Benchmark Reference:** Helsinki Kalasatama, Singapore Forward / CrowdTaskSG
- **Outcome Dimensions Supported:** Service relevance; Policy responsiveness; Trust and legitimacy; Quality of place-based outcomes



Inclusive service access through multilingual digital capability

- Ensure government services and information are accessible across the full linguistic diversity of the resident population, reducing participation barriers and strengthening trust across demographic groups. This could include:
 - Providing core government services and information in languages reflecting actual population composition
 - Deploying AI-powered real-time translation for service interactions
 - Ensuring multilingual capability extends beyond websites to call centers, chatbots and in-person touchpoints
 - Proactive outreach in community languages rather than relying on residents to seek translated content
- The aim is to close the gap between service availability and actual access for residents who speak neither Arabic nor English as a first language, strengthening inclusion across the city
- **Benchmark Reference:** Seoul 120 Dasan Call Centre, Helsinki InfoFinland, Zurich Cantonal Integration Programmes
- **Outcome Dimensions Supported:** Social inclusion; Service accessibility; Resident trust; Integration outcomes



Social Adaptivity



Place-based social cohesion through integrated civic anchors

- Creating physical spaces that bring together services, community programming and civic participation in ways that foster interaction across demographic groups. This could involve:
 - Co-locating government services, libraries, community spaces and recreational facilities in single integrated hubs
 - Designing spaces that encourage social integration across income levels, nationalities and age groups and active programming that creates reasons for community interaction.
 - Embedding feedback and participation mechanisms into the physical infrastructure.
- **Benchmark Reference:** Singapore Our Tampines Hub, Helsinki Oodi
- **Outcome Dimensions Supported:** Social cohesion; Community Identity; Service accessibility



Environmental Adaptivity

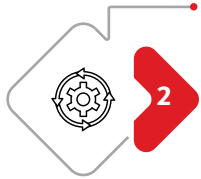


Demand-side energy activation through behavioral incentive programmes

- Complementing supply-side clean energy investment with mass-participation programmes that incentivize household and business energy reduction through verified savings and redeemable rewards. This could involve:
 - Smart meter integration enabling verified consumption tracking.
 - Points / rewards systems for measurable reduction against baselines.
 - Gamification and community competition elements.
 - Redemption pathways through public transport, utilities, or retail partners.
- **Benchmark Reference:** Seoul Eco-Mileage
- **Outcome Dimensions Supported:** Energy efficiency; Behavioral sustainability; Resident engagement; Demand management



Environmental Adaptivity



Construction circularity through systematic material recovery and reuse

- Embed circularity within the construction and demolition cycle by shifting from disposal-led practices toward systematic material recovery, tracking and reintegration into construction supply chains, in order to retain material value, reduce embodied carbon and improve resource efficiency at scale. This can involve:
 - Mandatory pre-demolition audits identifying recoverable materials and reuse potential
 - Material passports documenting composition, quality and reuse history to reduce liability and quality uncertainty
 - Market mechanisms connecting demolition outputs with construction demand
 - Alignment of permitting and compliance processes so reuse and deconstruction operate as viable alternatives to disposal
- **Benchmark Reference:** Helsinki mandatory pre-demolition audits; Singapore BCA material passports
- **Outcome Dimensions Supported:** Circular economy; Resource efficiency; Embodied-carbon reduction

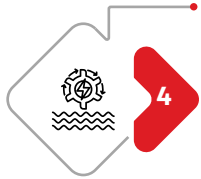


Resident and commercial reuse systems to reduce high-volume waste

- Reduce high-volume resident and commercial waste by strengthening reuse, repair and early interception systems across households, offices, retail and hospitality, reframing waste as a shared resource. Intervene downstream, at the point of disposal, by redesigning municipal waste and collection systems to intercept reusable materials before landfill or incineration and by strengthening secondary demand across communities and businesses.. This can involve:
 - Reuse and repair hubs serving residents, offices, retail and hospitality
 - Early interception mechanisms for furniture, appliances, fixtures and equipment prior to disposal
 - Service design and incentives that encourage reuse, donation, refurbishment and repair across residential and commercial premises
 - Structured participation mechanisms engaging property managers, landlords and businesses alongside residents
- **Benchmark Reference:** Tallinn (city-integrated reuse and circular economy centers)
- **Outcome Dimensions Supported:** Waste reduction; Resource efficiency; Circular economy; Community and business participation; Landfill diversion



Environmental Adaptivity



Strengthen Waste-Water-Energy nexus governance to maximize resource recovery

- Strengthen environmental adaptivity by integrating planning, performance objectives and operational coordination across waste, water and energy systems, so that resource recovery and system optimization are maximized across infrastructure. Co-location, shared process design and coordinated operations enable waste streams to be dynamically redirected toward energy and resource recovery, increasing system resilience and efficiency.
- Advancing this approach would require:
 - Formalized cross-system planning and optimization mechanisms linking water, energy and waste functions, where responsibilities span DEWA and municipal or project-based waste operations, with shared outcome metrics
 - Early integration of nexus considerations into greenfield districts and major brownfield expansions, rather than relying on post-hoc retrofitting
 - Clear accountability arrangements to resolve cross-utility trade-offs where optimization for one system affects others
- **Benchmark Reference:** Singapore (Tuas Nexus); Denmark (Billund BioRefinery)
- **Outcome Dimensions Supported:** Resource efficiency; energy resilience; environmental sustainability; infrastructure optimization; system-level adaptivity



Formalize nature-based solutions as functional urban infrastructure

- Strengthen climate and environmental adaptivity by explicitly defining nature-based solutions as functional urban infrastructure within planning, zoning and infrastructure frameworks, with clear performance roles linked to cooling, airflow, erosion control and biodiversity connectivity.
- This can involve:
 - Explicit articulation of functional roles for nature-based assets within planning and zoning instruments (e.g., cooling, airflow, erosion control, habitat connectivity)
 - Integration of nature-based performance objectives into infrastructure and district-level planning, alongside engineered solutions
 - Clear long-term management and accountability arrangements to ensure performance is maintained over time
 - Use of data and modelling (microclimate, airflow, coastal dynamics, biodiversity corridors) to guide placement and design
- **Benchmark Reference:** Seoul (Baram-gil Sup Wind Path Forests); Tallinn (Pollinator Highway)
- **Outcome Dimensions Supported:** Climate resilience; urban heat mitigation; biodiversity protection; environmental sustainability; system-level adaptivity



Urban Systems Adaptivity



Strengthen network-level reliability in conjunction with climate-mitigated transfers

- Strengthen urban mobility adaptivity by explicitly embedding network-level reliability and transfer performance as core objectives within mobility planning and operations, in conjunction with selective climate-mitigated transfer environments, so that network expansion remains usable under extreme heat and evolving travel patterns. This can involve:. This could involve:
 - Explicit articulation of network-level performance roles within mobility planning instruments (e.g., transfer reliability, recovery time, missed-connection thresholds), alongside corridor-level metrics
 - Integration of reliability and transfer-performance objectives into service planning, timetable coordination and operational control frameworks
 - Targeted climate-mitigation requirements at priority transfer locations, based on footfall, transfer distance and network criticality, rather than citywide walkability assumptions
 - Periodic reassessment of priority nodes and mitigation needs using operational data, as new districts are added and travel patterns shift
- **Benchmark Reference:** Zurich (network-wide transfer coordination); Singapore (Walk2Ride, selective climate-mitigated access)
- **Outcome Dimensions Supported:** Network resilience; Public transport usability; Climate-responsive mobility; System-level adaptivity



Differentiate first-last mile mobility strategies by district typology

- Strengthen urban mobility adaptivity by explicitly differentiating first-last mile strategies by district type, with the understanding that varying densities, land-use mixes and climate exposure require different mobility treatments rather than uniform citywide solutions. This could involve:
 - Formal classification of district typologies within mobility and land-use planning frameworks (e.g., high-density TODs, mixed-use centers, low-density residential areas, employment zones)
 - Definition of appropriate first-last mile approaches for each typology, such as enclosed or climate-mitigated pedestrian links, feeder services, park-and-ride integration, or micro-mobility where viable
 - Integration of first-last mile requirements into district planning and development controls, ensuring solutions are embedded early rather than retrofitted
 - Periodic review and adjustment of district classifications and treatments based on observed usage patterns, demand shifts and network expansion
- **Benchmark Reference:** Walk2Ride Programme & Integrated Climate-Controlled Links
- **Outcome Dimensions Supported:** Public transport accessibility; Climate-responsive mobility; Efficient network expansion; System-level adaptivity

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Thank You